Spatio-Temporal Air Quality Forecasting in Urban Sri Lanka: A Comparative Multi-Modal Analysis of Colombo and Kandy

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List of Abbreviations

AQI: Air Quality Index

ARIMA: Autoregressive Integrated Moving Average

BAM: Beta Attenuation Monitor

BCS: British Computer Society

CEA: Central Environmental Authority

CTM: Chemical Transport Models

DDoS: Distributed Denial-of-Service

DL: Deep Learning

ESA: European Space Agency

FCM: Firebase Cloud Messaging

GDPR: General Data Protection Regulation

GEE: Google Earth Engine

LESP: Legal, Ethical, Social, and Professional

LSTM: Long Short-Term Memory

MAE: Mean Absolute Error

ML: Machine Learning

NO2: Nitrogen Dioxide

PDPA: Personal Data Protection Act

PM2.5: Fine Particulate Matter

PoC: Proof of Concept

RF: Random Forest

RMSE: Root Mean Squared Error

RNN: Recurrent Neural Networks

SARIMA: Seasonal Autoregressive Integrated Moving Average

SARIMAX: Seasonal Autoregressive Integrated Moving Average with eXogenous factors

SHAP: SHapley Additive exPlanations

TROPOMI: Tropospheric Monitoring Instrument

UTC: Coordinated Universal Time

WHO: World Health Organization

XAI: Explainable AI

XGBoost: Extreme Gradient Boosting

CHAPTER 1: INTRODUCTION

1.1 Chapter Overview

This chapter establishes the foundational context for the proposed research, outlining the critical need for advanced spatio-temporal air quality forecasting in urban Sri Lanka. It transitions from a broad overview of global atmospheric pollution challenges to the localized crisis facing the cities of Colombo and Kandy. By defining the limitations of current reactive monitoring systems and exploring the gaps in existing literature, this chapter formulates the core problem statement. Subsequently, it delineates the specific research aims, actionable objectives, and the overarching scope of the study, ultimately providing a clear roadmap for the comparative multi-modal analysis undertaken in this project.

1.2 Background and Problem Domain

Air pollution is universally recognized as one of the most severe environmental threats to public health. According to the World Health Organization (WHO), prolonged exposure to fine particulate matter (PM_2.5) is intrinsically linked to heightened incidences of cardiovascular diseases, respiratory illnesses, and premature mortality. While developed nations have heavily invested in dense, state-of-the-art environmental monitoring grids, developing nations within South Asia frequently struggle to maintain adequate infrastructure. Sri Lanka, a rapidly urbanizing island nation, finds itself at a critical juncture regarding environmental management, where increasing vehicular emissions, industrial expansion, and distinct topographical challenges exacerbate local air quality degradation (Rathnayake, Sakurai and Weerasekara, 2023).

1.2.1 Context: The Tale of Two Cities

The atmospheric dynamics of Sri Lanka are highly heterogeneous, heavily influenced by seasonal monsoons and localized topography. This research focuses on two socio-economically vital but geographically distinct urban centers: Colombo and Kandy. Colombo, the commercial capital situated on the western coastline, experiences high volumes of traffic-induced pollution that is frequently dispersed by coastal wind patterns. In stark contrast, Kandy is nestled within a valley in the central highlands. This unique bowl-shaped topography severely restricts wind circulation, leading to severe atmospheric inversions where pollutants—particularly localized vehicular exhaust and biomass burning—become trapped at ground level for extended periods. Consequently, despite having a smaller population and less industrialization than Colombo, Kandy frequently records (PM_2.5) concentrations that surpass WHO safety guidelines, making it a critical subject for localized environmental modeling.

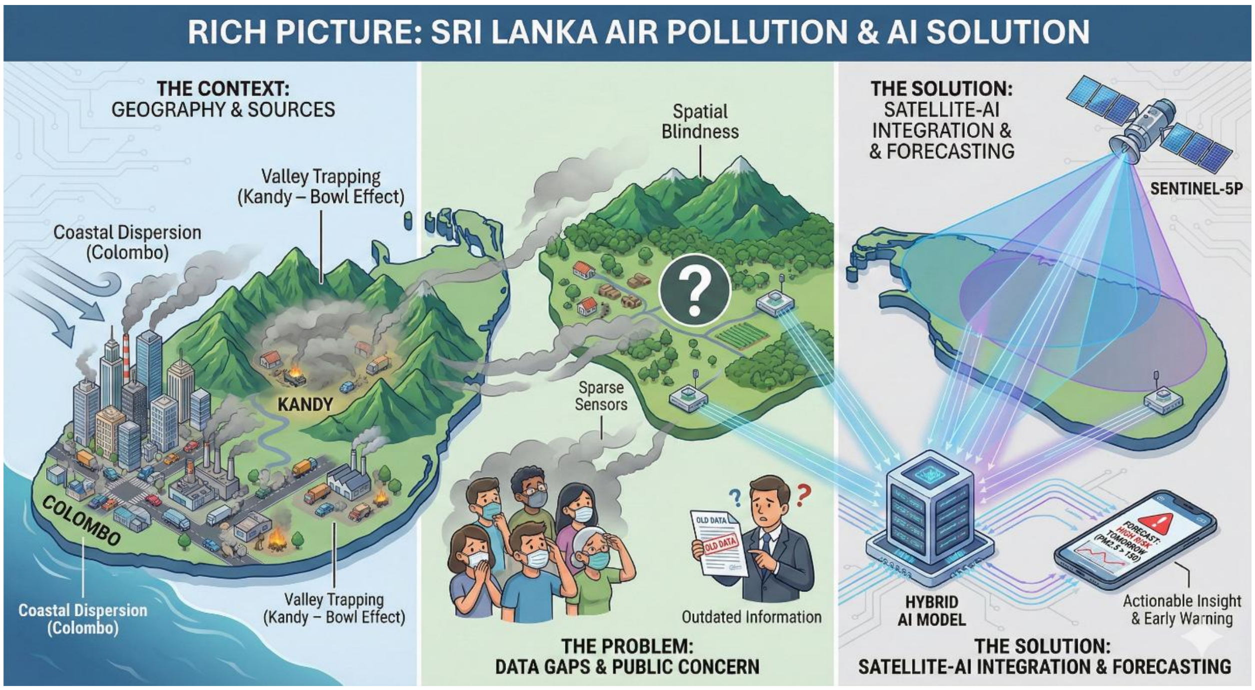

Figure 1 Rich picture

1.2.2 The Limitation of Retrospective Monitoring

Historically, Sri Lanka's approach to air quality management has been predominantly retrospective. Relying on sporadic government monitoring stations and localized low-cost sensor networks (such as PurpleAir), authorities can only react to pollution spikes *after* they have occurred. While real-time dashboards provide immediate situational awareness, they fail to offer the predictive foresight necessary for proactive public health interventions. For instance, advising vulnerable populations to remain indoors or implementing temporary traffic restrictions requires a reliable forecast of at least 24 to 48 hours.

1.2.3 The Monitoring Gap: Ground Sensors vs. Satellite Telemetry

A secondary systemic issue within the problem domain is the physical vulnerability of ground-based sensor networks. Low-cost sensors are susceptible to power outages, Wi-Fi disconnections, and hardware degradation, leading to significant temporal gaps in historical datasets. Conversely, satellite telemetry—such as the Tropospheric Nitrogen Dioxide (NO_2) column density captured by the European Space Agency's Sentinel-5P satellite—provides uninterrupted, macro-level atmospheric data. However, satellite data alone lacks the micro-level granularity required to accurately determine the exact (PM_2.5) concentration at street level. Therefore, a significant computational challenge exists in bridging the macro-scale reliability of satellite imagery with the micro-scale precision of ground sensors to create a resilient, unified predictive model (Rowley and Karakuş, 2023).

1.3 Problem Definition & Statement

Despite the escalating threat of atmospheric pollution in Sri Lanka, the current infrastructure relies heavily on reactive, ground-level monitoring that is highly susceptible to mechanical and infrastructural failure. When physical sensors go offline—sometimes for months at a time—critical data continuity is lost, rendering traditional time-series forecasting models ineffective. Furthermore, existing forecasting attempts in the region typically rely on single-modality datasets (either exclusively ground data or exclusively satellite data), limiting their predictive accuracy.

Therefore, the core problem this research addresses is the lack of a robust, multi-modal forecasting framework capable of proactively predicting urban (PM_2.5) concentrations while remaining resilient to the inherent data gaps found in developing nations' ground-sensor networks.

1.4 The Research Gap

A preliminary review of contemporary literature reveals a significant dichotomy in air quality modeling. Studies utilizing traditional statistical methods (such as SARIMA) often struggle to capture the complex, non-linear spikes associated with urban emissions (Mampitiya et al., 2023). Conversely, while advanced Machine Learning (ML) and Deep Learning (DL) models are increasingly popular, they are notoriously data-hungry and fail catastrophically when presented with prolonged missing sequences—a common occurrence in Sri Lankan sensor networks.

Crucially, there is a distinct lack of comparative research focusing on the *fusion* of macro-level satellite telemetry (specifically Sentinel-5P Tropospheric NO_2) with micro-level ground sensors (PurpleAir) across contrasting Sri Lankan topographies. Furthermore, the application of satellite data not just as an exogenous forecasting variable, but as a mechanism for *imputing* catastrophic ground-sensor blackouts, remains a largely unexplored frontier in South Asian environmental data science.

1.5 Research Aim and Objectives

1.5.1 Research Aim

The primary aim of this research is to design, implement, and evaluate a multi-modal spatio-temporal air quality forecasting framework that fuses satellite (NO_2) telemetry with ground-based (PM_2.5) sensor networks. By comparing statistical, tree-based machine learning, and sequential deep learning architectures, this study seeks to determine the optimal methodology for predicting localized air quality across the distinct urban topographies of Colombo and Kandy.

1.5.2 Research Objectives

To achieve the overarching aim, the following actionable objectives have been established:

1. **Data Acquisition & Preprocessing:** To systematically extract, clean, and align four years (2022–2025) of hourly (PM_2.5) ground sensor data with daily Sentinel-5P (NO_2) satellite imagery for Colombo and Kandy.

2. **Multi-Modal Data Fusion:** To engineer a cohesive temporal dataset that maps low-frequency satellite observations to high-frequency ground sensor readings.

3. **Baseline Statistical Modeling:** To develop and evaluate a SARIMAX model to serve as the traditional statistical baseline for predictive accuracy.

4. **Advanced Predictive Modeling:** To design, train, and hyper-parameterize both a Tree-Based Machine Learning architecture (Random Forest/XGBoost) and a Deep Learning architecture (Long Short-Term Memory - LSTM) for short-term pollution forecasting.

5. **Algorithmic Evaluation:** To critically compare the performance of the implemented models using standard error metrics (RMSE and MAE) and Feature Importance analysis (SHAP).

6. **Satellite Imputation (Advanced Objective):** To utilize satellite (NO_2) telemetry as a proxy variable to mathematically impute and reconstruct prolonged periods of missing

ground sensor data (specifically addressing the 6-month sensor failure in the Kandy dataset).

1.6 Research Questions

This study is driven by the following core research questions:

RQ1: To what extent can the computational fusion of daily satellite chemical vectors (NO_2) with hourly ground data improve the RMSE of PM_2.5 forecasts in distinct urban topographies (Colombo vs. Kandy)?

RQ2: Does a Hybrid LSTM-Dense architecture demonstrate statistically significant superiority over traditional machine learning models (XGBoost) in capturing non-linear pollution trends in valley (Kandy) and coastal (Colombo) environments?

RQ3: What is the quantifiable variance in PM_2.5 levels attributed specifically to vehicular emissions NO_2 versus meteorological dispersion in these two cities, as isolated by SHAP analysis?

1.7 Scope and Limitations

1.7.1 Scope of the Study

The spatial scope of this research is strictly defined to the urban centers of Colombo (coastal) and Kandy (valley). The temporal scope spans a continuous four-year period from January 1, 2022, to December 31, 2025. The technological scope encompasses data extraction via the PurpleAir API and Google Earth Engine (GEE), followed by model development utilizing Python-based libraries (Scikit-Learn, TensorFlow/Keras, and Statsmodels). The primary target variable is strictly limited to (PM_2.5), utilizing NO_2, temperature, and humidity as predictive features.

1.7.2 Limitations

This study acknowledges several inherent limitations. Firstly, Sentinel-5P optical satellite sensors cannot penetrate heavy cloud cover, resulting in missing (NO_2) observations during Sri Lanka's intense monsoon seasons; these gaps necessitate linear interpolation which may slightly reduce micro-level accuracy. Secondly, computational constraints limit the ability to train Deep Learning models across massive multi-year datasets without aggregation or down-sampling. Finally, the study relies on the continuous uptime of third-party APIs (PurpleAir), meaning hardware degradation out of the researcher's control may dictate the volume of available training data.

1.8 Chapter Summary

This chapter introduced the critical necessity for predictive air quality modeling in Sri Lanka, highlighting the vulnerabilities of relying solely on physical ground sensors. By identifying the research gap surrounding multi-modal satellite fusion and establishing a clear set of objectives, the framework for a robust comparative study has been laid. The subsequent chapter will critically review existing literature to evaluate how previous researchers have approached spatio-temporal forecasting, thereby informing the specific methodological choices made in this project.

CHAPTER 2: LITERATURE REVIEW

2.1 Chapter Overview

This chapter presents a critical, thematic review of the existing literature surrounding air quality forecasting, with a specific focus on urban particulate matter (PM_2.5). The review traces the chronological and methodological evolution of environmental modeling, transitioning from early deterministic physical models to modern, data-driven computational architectures. By systematically evaluating traditional statistical methods (SARIMA), Tree-Based Machine Learning (Random Forest/XGBoost), and Deep Learning frameworks (LSTM), this chapter establishes the theoretical foundation for the methodologies chosen in this study. Furthermore, it critically examines the emerging paradigm of multi-modal data fusion—specifically the integration of satellite telemetry with ground sensors—to explicitly highlight the research gaps that this project seeks to fulfill within the Sri Lankan context.

2.2 The Evolution of Air Quality Forecasting

The prediction of atmospheric pollutants has historically been a complex challenge due to the chaotic, non-linear interactions between meteorological variables, topographical features, and anthropogenic emissions. Early forecasting methodologies relied heavily on Chemical Transport Models (CTMs) and deterministic dispersion models (Liang et al., 2020). While these physical models provided valuable insights into the chemical life-cycles of pollutants, they were computationally exhaustive, required massive arrays of micro-meteorological input data, and often struggled to provide accurate short-term, localized forecasts.

As computational power increased, the paradigm shifted toward data-driven, empirical modeling. Rathnayake, Sakurai and Weerasekara (2023) conducted a comprehensive review of outdoor air pollution in Sri Lanka compared to the wider South Asian region, noting that while neighboring countries like India have heavily adopted data-driven forecasting networks, Sri Lanka has lagged. The authors highlight that Sri Lanka's unique topography—particularly the contrast between the coastal wind-swept plains of Colombo and the inversion-prone highland valleys of Kandy—renders generalized regional forecasting models highly inaccurate. Consequently, modern literature dictates that effective forecasting in such heterogeneous environments requires localized, high-frequency data modeling. This consensus directly justifies this project's comparative approach, necessitating models that can learn the distinct, localized patterns of both Colombo and Kandy rather than applying a universal algorithm.

2.3 Traditional Statistical Methods (SARIMA/ARIMA)

Before the widespread adoption of Machine Learning, stochastic time-series forecasting dominated environmental data science. The Autoregressive Integrated Moving Average (ARIMA) model, and its seasonal extension (SARIMA), have been extensively utilized to predict (PM_2.5) concentrations. SARIMA models operate on the premise of linear correlation, forecasting future values based on a linear combination of past observations (autoregression) and past forecasting errors (moving average), while accounting for periodic fluctuations (seasonality).

In a recent comparative study within the Sri Lankan context, Mampitiya et al. (2023) evaluated the efficacy of traditional models against emerging machine learning techniques using meteorological data in urban areas. The consensus in the literature is that while SARIMA serves as an excellent, highly interpretable baseline, it possesses fatal mathematical limitations when applied to modern environmental datasets. First, SARIMA fundamentally assumes that the underlying data generating process is stationary and linear. However, as noted by Zhu et al. (2018), urban (PM_2.5) generation is inherently non-linear, driven by sudden anomalies such as abrupt traffic congestion, localized biomass burning, or sudden monsoon downpours. When presented with these chaotic spikes, SARIMA's linear mathematical constraints force it to predict a smoothed "average" trajectory, completely failing to capture the extreme pollution events that pose the highest risk to public health.

Furthermore, traditional statistical models struggle significantly with high-frequency dimensional scaling. To capture an annual seasonal trend using hourly sensor data, a SARIMA model requires a lag interval of 8,760 time-steps, a computation that is often mathematically prohibitive and leads to severe model overfitting. This limitation was echoed in the *Proceedings of the International Research Conference of the Open University of Sri Lanka* (2023), where comparative analyses demonstrated that while SARIMA models perform adequately on aggregated daily or monthly averages, their accuracy degrades catastrophically when applied to high-frequency, hour-by-hour predictions. Therefore, while this project utilizes SARIMAX (accounting for exogenous variables) to establish a rigorous baseline, the literature strongly indicates that satisfying the demand for highly accurate, hourly urban forecasts requires a transition to advanced algorithmic architectures.

2.4 Machine Learning Approaches: Tree-Based Architectures

To overcome the non-linear limitations of traditional statistical models, environmental data scientists have increasingly adopted ensemble Machine Learning (ML) architectures, predominantly Random Forest (RF) and Extreme Gradient Boosting (XGBoost). Mihirani and Yasakethu (2023) highlight the growing efficacy of machine learning-based air pollution prediction models, noting their superior ability to map complex, multidimensional interactions between meteorological variables (temperature, humidity) and pollutant concentrations.

Tree-based models operate by constructing multitudes of decision trees during training and outputting the mean prediction of the individual trees. Zhu et al. (2018) emphasize that the primary advantage of these models in air quality prediction lies in their inherent robustness to outliers and their lack of rigid assumptions regarding data distribution. In the context of Sri Lankan urban forecasting, where a sudden localized fire or sudden traffic gridlock might create a massive, anomalous PM_2.5 spike, Tree-based models can isolate these anomalies into specific decision branches without skewing the overall mathematical trend.

However, the literature also identifies a critical vulnerability in standard ML approaches: the "Lag-1 Autocorrelation Paradox." Tree-based algorithms often rely so heavily on the immediate historical value (the PM_2.5 level one hour prior) that they struggle to project accurate forecasts beyond a 1-to-3 hour horizon. Furthermore, while highly accurate, ensemble trees are traditionally "black-box" algorithms. To satisfy the demands of transparent environmental policy-making, modern literature mandates the integration of explainable AI techniques. Tools such as SHAP (SHapley Additive exPlanations) are now required to deconstruct model outputs, quantifying exactly how secondary variables like humidity or vehicular emissions directionally influence the final PM_2.5 prediction.

2.5 Deep Learning and Sequential Modeling (LSTM)

While Tree-based ML excels at tabular feature mapping, it does not possess an inherent, mathematical understanding of sequential time. To address the continuous, temporal nature of atmospheric cycles, researchers have pivoted towards Deep Learning (DL), specifically Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. Liang et al. (2020) comprehensively explored machine learning-based predictions of air quality, demonstrating that deep sequential architectures consistently outperform both SARIMA and standard ML when forecasting across longer multi-horizon windows (e.g., 24 to 48 hours).

The architectural superiority of the LSTM lies in its specialized memory cells, gated by input, output, and forget mechanisms. This allows the network to learn both short-term micro-fluctuations (e.g., daily rush hour spikes) and long-term macro-dependencies (e.g., seasonal monsoon shifts) simultaneously. By analyzing a continuous "lookback" window, LSTMs dynamically update their internal state, theoretically making them the ultimate forecasting tool for urban pollution.

Despite this, the literature outlines a severe operational constraint: Deep Learning architectures are notoriously data-hungry and fragile when confronted with discontinuous datasets. If a physical sensor network experiences a catastrophic power failure, resulting in prolonged data blackouts, the LSTM's sequential memory chain is broken. This operational reality exposes a critical flaw in applying DL to developing nations where sensor infrastructure is unreliable, necessitating a methodological shift toward data fusion and algorithmic imputation to ensure model stability.

2.6 Multi-Modal Data Fusion and Satellite Imputation

The most promising frontier in contemporary environmental data science is multi-modal data fusion—the integration of micro-level ground sensor telemetry with macro-level satellite observations. Rowley and Karakuş (2023) pioneered frameworks for predicting air quality via multimodal AI and satellite imagery, explicitly demonstrating that fusing spatial and temporal data yields predictive accuracies unattainable by single-modality models.

Satellites such as the European Space Agency's Sentinel-5 Precursor (Sentinel-5P) provide continuous, daily global coverage of trace gases, most notably Tropospheric Nitrogen Dioxide (NO_2). Because NO_2 is a primary byproduct of fossil fuel combustion, it serves as a highly reliable, leading atmospheric indicator of urban (PM_2.5) generation. While ground sensors are vulnerable to localized hardware failures, satellite telemetry remains structurally immune to ground-level infrastructural collapse.

Currently, the majority of literature utilizes satellite data strictly as an exogenous feature to enhance standard forecasting models. However, a significant, unexplored opportunity exists in utilizing this multi-modal relationship for algorithmic imputation. If a robust mathematical correlation between atmospheric NO_2 and ground-level PM_2.5 can be established using Machine Learning, satellite data could theoretically be used as a proxy to retrospectively reconstruct massive gaps in historical ground sensor data. This application of satellite fusion represents a profound evolution from simple predictive modeling to automated infrastructural resilience.

2.7 Summary of Identified Research Gaps

Based on the critical thematic review of the current literature, this study identifies three distinct research gaps that must be addressed within the context of South Asian environmental modeling:

1. **The Single-Modality Limitation:** Most Sri Lankan air quality studies rely exclusively on either isolated ground sensors or broad satellite observations, failing to harness the synergistic predictive power of multi-modal data fusion.
2. **Topographical Neglect:** Existing models are frequently generalized at the provincial or national level. There is a distinct lack of comparative analyses evaluating how specific machine learning architectures perform when transitioned from coastal, wind-swept topographies (Colombo) to highly restrictive, inversion-prone highland valleys (Kandy).
3. **The Data Continuity Crisis:** Current forecasting literature assumes the availability of continuous, unbroken temporal datasets. There is a critical void in research addressing how to utilize satellite proxy data (such as Sentinel-5P NO_2) to mathematically impute and salvage catastrophic, multi-month data blackouts in urban ground sensor networks.

This Final Year Project is explicitly designed to bridge these three gaps, proposing a multi-modal, comparative framework that not only forecasts air quality but actively addresses the infrastructural realities of data collection in developing urban environments.

CHAPTER 3: METHODOLOGY

3.1 Chapter Overview

This chapter delineates the structured methodology employed to achieve the research aims and objectives outlined in Chapter 1, while directly addressing the computational gaps identified in the

Chapter 2 literature review. Utilizing Saunders' Research Onion as a structural framework, this section justifies the chosen research philosophy and methodological approach. Subsequently, it details the high-level system architecture, the strategy for multi-modal data acquisition, and the preprocessing pipelines required to fuse high-frequency ground sensor data with low-frequency satellite telemetry. Finally, the chapter defines the mathematical evaluation metrics used to benchmark the algorithms and outlines the project management constraints, including risk mitigation strategies.

3.2 Research Philosophy and Approach

To ensure rigorous academic validity, the methodological design of this study is grounded in the outer layers of Saunders' Research Onion, specifically defining the research philosophy and approach.

3.2.1 Positivist Research Philosophy

This study adopts a strictly **Positivist** research philosophy. Positivism asserts that authentic knowledge is derived from the observation of objective, measurable phenomena, independent of human bias or subjective interpretation. Because this research involves the extraction, processing, and mathematical modeling of quantifiable atmospheric variables (specifically PM_2.5, NO_2, temperature, and humidity) across four years, a positivist paradigm is the only appropriate philosophical stance. The researcher acts as an objective observer, utilizing statistical and computational algorithms to uncover patterns and establish causal relationships within the atmospheric data.

3.2.2 Deductive Research Approach

In alignment with the positivist philosophy, this project utilizes a **Deductive** research approach. Rather than generating new theories from raw observations (inductive), this study begins with established hypotheses derived from existing literature: primarily, the hypothesis that fusing satellite NO_2 telemetry with ground sensor data will mathematically improve the predictive accuracy of urban PM_2.5 forecasting algorithms. The methodology is designed to empirically test this hypothesis by building, training, and comparatively evaluating specific computational architectures (SARIMAX, Tree-based ML, and LSTM) against unseen test datasets.

3.3 System Architecture and Design Strategy

The core of this research relies on a complex, multi-modal computational pipeline designed to ingest, align, and model disparate environmental datasets. The system architecture is logically divided into four distinct sequential phases: Data Acquisition, Preprocessing & Fusion, Algorithmic Modeling, and Comparative Evaluation.

3.3.1 Phase 1: Multi-Modal Data Acquisition

The architecture features dual-ingestion pipelines. The ground-level pipeline utilizes RESTful API requests to extract high-frequency (hourly) PM_2.5 and meteorological data from physically deployed PurpleAir sensor nodes in Colombo and Kandy. Concurrently, the macro-level pipeline utilizes the Google Earth Engine (GEE) JavaScript API to extract low-frequency (daily) Tropospheric NO_2 column density from the Copernicus Sentinel-5P satellite, precisely filtered to the spatial coordinates of the respective ground sensors.

3.3.2 Phase 2: Preprocessing and Temporal Fusion

Because the two data modalities operate on fundamentally different temporal frequencies (hourly vs. daily), the architecture requires a robust temporal alignment module. The preprocessing engine standardizes all timestamps to Sri Lanka Standard Time (UTC +5:30) and forces the data into a strict one-hour chronometric grid. Mathematical interpolation techniques (linear and polynomial) are employed to handle minor sensor disconnections, while historical rolling averages are utilized for longer operational gaps. Finally, a Left-Join fusion protocol is executed, broadcasting the single daily satellite NO_2 reading across the corresponding 24-hour ground sensor grid, resulting in a cohesive, multi-modal training matrix.

3.3.3 Phase 3: Algorithmic Modeling

The processed matrices are split into chronological training (2022–2024) and testing (2025) datasets. The architecture evaluates three distinct computational models:

1. **The Statistical Baseline:** A SARIMAX (Seasonal Autoregressive Integrated Moving Average with eXogenous factors) model, serving to evaluate linear, seasonal predictions based on aggregated daily data.

2. **Tree-Based Machine Learning:** An ensemble Random Forest/XGBoost regressor, utilizing complex engineered features (such as 24-hour lagging variables and rolling means) to capture non-linear, short-term urban spikes.

3. **Sequential Deep Learning:** A Long Short-Term Memory (LSTM) neural network, leveraging a 24-hour sequential 'lookback' window to mathematically map both long-term weather dependencies and short-term pollution anomalies.

3.3.4 Phase 4: Advanced Satellite Imputation (Kandy Strategy)

A unique architectural sub-routine is designed specifically to address the infrastructural reality of developing nations: massive sensor blackouts. For the Kandy dataset, which experienced a catastrophic 6-month hardware failure, the architecture employs the trained Tree-Based ML model in reverse. By feeding the unbroken Sentinel-5P satellite NO_2 data into the model during the blackout period, the architecture mathematically imputes and reconstructs the missing ground-level PM_2.5 values, effectively salvaging the historical dataset for future forecasting.

3.4 Data Acquisition Strategy and Technical Specifications

To implement a truly multi-modal predictive framework, this study relies on data streams retrieved from two distinct environmental tracking systems. By executing targeted requests across these networks, a synchronized dataset spanning from January 1, 2022, to December 31, 2025, was established.

3.4.1 Ground-Level Telemetry via PurpleAir REST API

Ground-level particulate matter and meteorology are collected from high-precision, low-cost optical laser counters maintained by the PurpleAir sensor network. The data extraction architecture executes HTTP GET requests against the PurpleAir History API endpoint (https://api.purpleair.com/v1/sensors/history/csv), authenticated via an encrypted Read API key.

To conduct a comparative spatial analysis, two target sensors were isolated based on their geographic positioning and data density:

1. **Colombo Sensor Node (Index: 29677):** Situated at spatial coordinates [79.9763° E, 6.9856° N]. This node captures the atmospheric profile of Sri Lanka's highly dense, industrialized coastal capital.

2. **Kandy Sensor Node (Index: 12451):** Situated at spatial coordinates [80.6337° E, 7.2906° N]. This node captures the microclimate of the central highland valley city, which is heavily prone to atmospheric inversion.

The API payload parameters are strictly configured to download data aggregated at a 60-minute interval (average=60), capturing a continuous stream of hourly averages. The parameters extracted include atmospheric fine particulate matter concentration (pm2.5_atm), (measured in $\mu g/m^3$), ambient temperature in Fahrenheit (temperature), and relative humidity percentage (humidity).

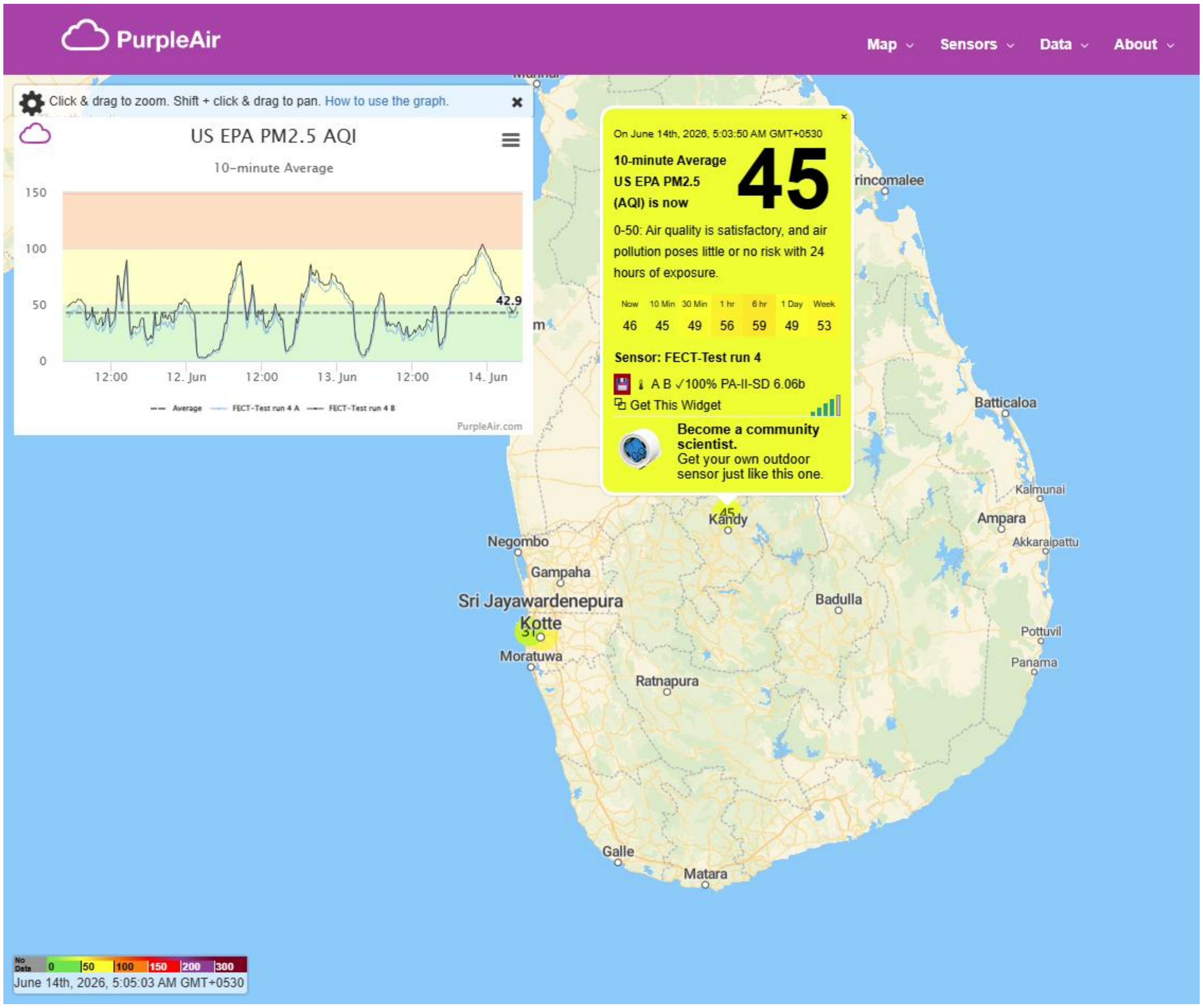

Figure 2 Purple Air website overview

3.4.2 Spaceborne Tropospheric Sensing via Google Earth Engine

Macro-level atmospheric gas mapping is achieved by extracting remote sensing telemetry from the Tropospheric Monitoring Instrument (TROPOMI) onboard the European Space Agency's (ESA) Sentinel-5 Precursor satellite. The extraction pipeline is implemented within the Google Earth Engine (GEE) cloud architecture, accessing the Earth Engine Data Catalog image collection ID: COPERNICUS/S5P/OFFL/L3_NO2.

The script isolates the Offline (OFFL) data product, which is rigorously reprocessed by ESA for historical research stability. The specific band selected is tropospheric_NO2_column_number_density, which quantifies the total number of Nitrogen

Dioxide molecules per unit area in a vertical column of the troposphere, expressed in moles per square meter mol/m².

The GEE engine executes a spatial reduction function (ee.Reducer.mean()) centered on the exact geographic points of the Colombo and Kandy ground sensors. The spatial scale is locked at 1113.2 meters to perfectly match the pixel resolution of TROPOMI at nadir. The resulting time-series captures a low-frequency, daily observation map that corresponds with the satellite's orbital revisit schedule over the Sri Lankan landmass.

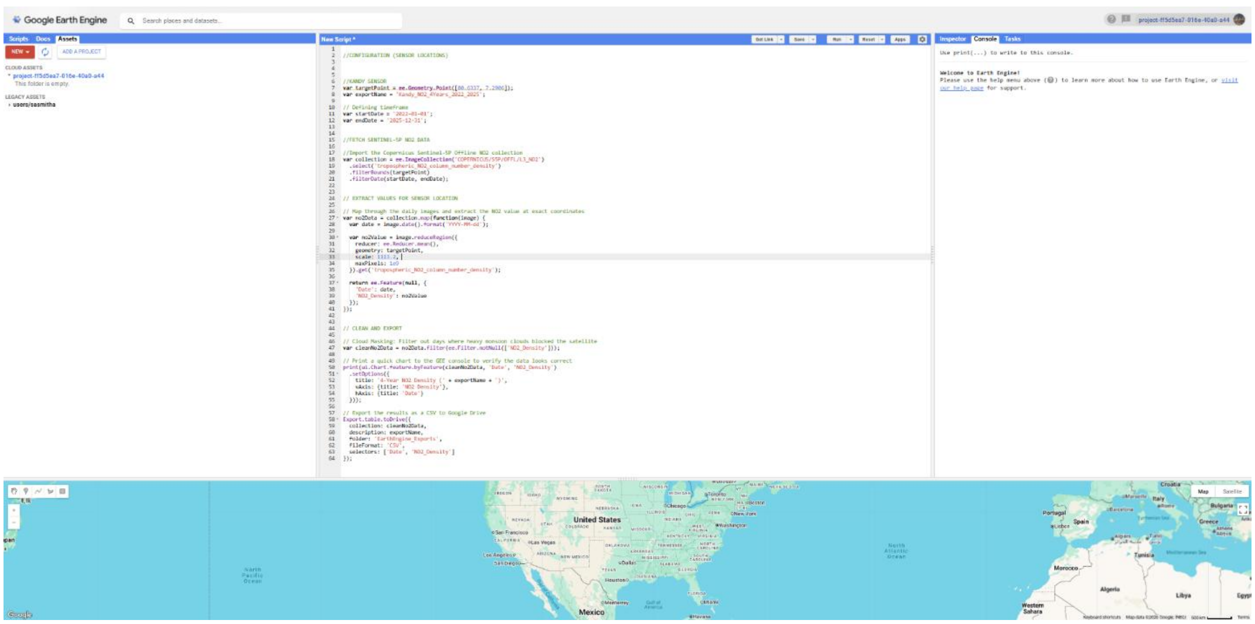

Figure 3 Google earth engine code editor

Parameter	**Source Type**	**Platform / Endpoint**	**Spatial Location / Core IDs**	**Temporal Frequency**	**Target Variables**
Ground Sensors	In-situ Optical Node	PurpleAir REST API	Colombo ID: 29677 Kandy ID: 12451	Hourly (60-min averages)	PM_2.5 (measured in $\mu g/m^3$), Temperature, Humidity
Satellite Imagery	Remote Sensing Multi-Spectral	Google Earth Engine Cloud	Copernicus Sentinel-5P TROPOMI	Daily Orbit	Tropospheric NO_2 column density mol/m²

Table 1 Sources table

3.5 Model Evaluation Metrics & Explainable AI Framework

To evaluate the predictive accuracy of the statistical, machine learning, and deep learning architectures, the models are evaluated on an unseen test dataset consisting of the full calendar year of 2025. Two standard mathematical loss functions are used to score accuracy, complemented by a cooperative game theory framework for feature interpretation.

3.5.1 Root Mean Squared Error (RMSE)

The primary metric used to quantify forecasting error variance is the Root Mean Squared Error. Mathematically, it is formulated as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Figure 4 RMSE formula

Where n represents the total number of hourly testing steps, y_i is the actual ground-truth PM_2.5 concentration recorded by the sensor, and $\hat{y}_i$ is the corresponding value predicted by the model. Because RMSE squares the differences before averaging, it inherently places a heavier mathematical penalty on larger errors. In the context of public health air quality forecasting, this characteristic is vital: a model that misses a catastrophic toxic smog spike by $40\ \mu g/m^3$ poses a significantly higher social risk than a model that makes multiple minor errors of $2\ \mu g/m^3$. Hence, minimizing RMSE ensures the models are structurally tuned to avoid critical underpredictions during acute pollution events.

3.5.2 Mean Absolute Error (MAE)

To complement the RMSE, the Mean Absolute Error is calculated to determine the linear, unweighted average magnitude of the forecasting errors. It is expressed as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Figure 5 MAE formula

MAE provides an intuitive, scale-dependent metric where every single error step is directly proportional to its absolute difference. This allows non-technical stakeholders and municipal authorities to easily interpret the model's baseline performance, establishing exactly how many micrograms per cubic meter ($\mu g/m^3$) the algorithm is off on any given hour.

3.5.3 Explainable AI (XAI) via SHAP

A common criticism of ensemble tree-based models and multi-layered deep learning networks is their inherent lack of transparency. To prevent these algorithms from operating as closed "black boxes," this methodology integrates **SHAP (SHapley Additive exPlanations)**. Based on cooperative game theory, SHAP calculates Shapley values to measure the specific marginal contribution of each input variable (NO_2, humidity, temperature, and historical lag variables) to the final PM_2.5 output. By deconstructing individual predictions, the SHAP framework visually maps the directional impact of features, demonstrating whether high levels of spaceborne NO_2 act as a positive accelerator for ground-level particulate matter across different urban topographies.

3.6 Project Management and Risk Mitigation

The execution of this research is governed by strict project management constraints to ensure the software engineering and data science pipelines are delivered systematically within the academic timeline.

3.6.1 Development Workflow

The project adopts an Iterative and Incremental development lifecycle. Rather than building the final complex model immediately, a scaled Proof of Concept (PoC) was developed using a restricted 3-month dataset from early 2023. This phase validated the REST extraction scripts, coordinate mapping, and basic LSTM compilation parameters. Upon successful validation, development scaled to the bulk 4-year phase. Progress is monitored via a dedicated Gantt chart, segmenting milestones into distinct sprints: Data Pipeline Engineering, Multi-Modal Fusion, Baseline Evaluation, Model Optimization, and Final Thesis Compilation.

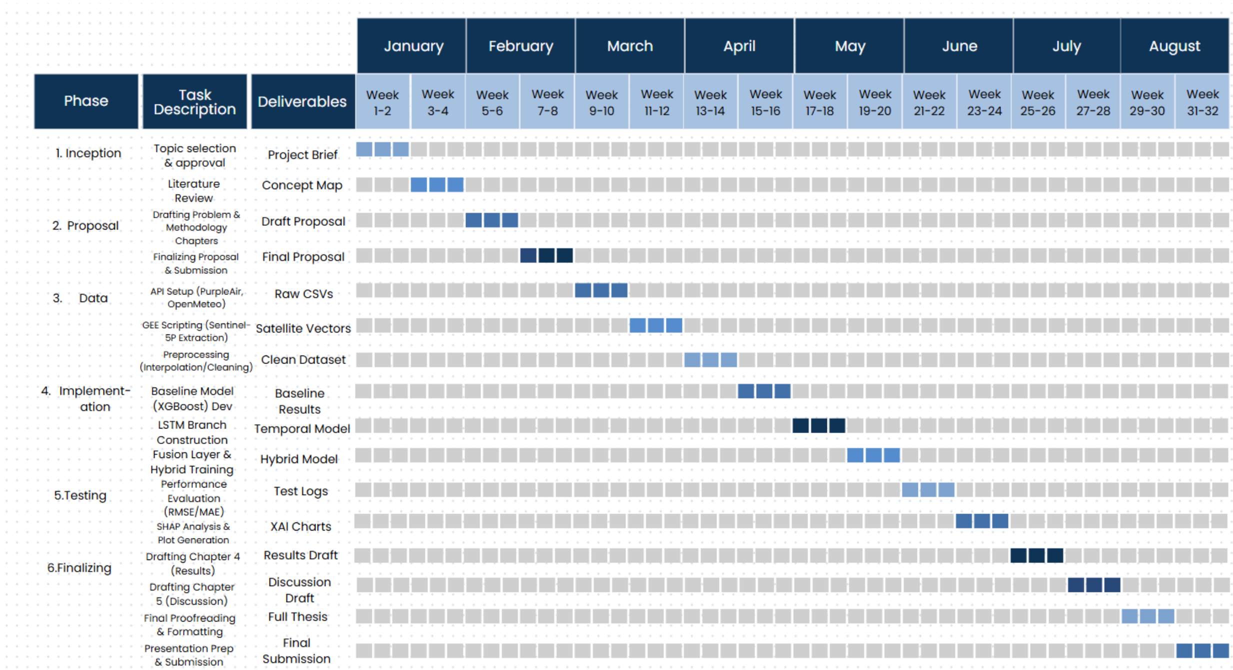

Figure 6 GANTT chart

To ensure the timely execution of the multi-modal data fusion and machine learning lifecycle, the project timeline was rigorously mapped using a Gantt chart framework. As illustrated in Figure

3.x below, the schedule accommodates iterative cycles of data acquisition, algorithmic training, and interface prototyping, allowing for built-in contingency buffers to handle API rate limits and unexpected sensor blackouts.

3.6.2 Risk Assessment and Mitigation Matrix

Working with third-party APIs, remote physical hardware, and satellite arrays introduces specific operational vulnerabilities. The matrix below outlines the anticipated project risks and their engineered computational mitigations:

Risk Description	**Impact Severity**	**Probability**	**Empirical Mitigation Strategy**
API Throttling & Payload Exceedance *(PurpleAir blocking bulk historical data requests)*	Medium	High	Implemented an automated monthly chunking loop inside the Python script, using time.sleep(2) rate-limiting to prevent server-side IP blocks.

Risk Description	Impact Severity	Probability	Empirical Mitigation Strategy
Catastrophic Hardware Blackouts *(Physical ground sensors losing power/Wi-Fi)*	High	Medium	Addressed Kandy's 6-month blackout by engineering a specific machine learning sub-routine that uses Sentinel-5P satellite data as a mathematical proxy to impute missing ground values.
Severe Cloud Cover Interference *(Satellites unable to map NO_2 due to monsoons)*	Medium	High	Integrated a dual-direction temporal linear interpolation module (limit_direction='both') inside the pandas fusion pipeline to smoothly bridge missing satellite days.
Local Environment Corruption	Low	Low	Maintained a locked, strict virtual python environment using requirements.txt to guarantee code execution portability between local systems and cloud interpreters.

Risk Description	**Impact Severity**	**Probability**	**Empirical Mitigation Strategy**
(Library dependency version mismatch)			

Table 2 Risk Assessment table

3.7 Chapter Summary

This chapter provided a comprehensive look at the research design of this project, establishing a positivist, deductive framework using Saunders' Research Onion. By clarifying the technical data parameters for Colombo and Kandy, detailing the mathematical formulations of your performance metrics, and establishing a clear risk mitigation framework, the execution path is fully solidified. The subsequent chapter will outline the legal, ethical, and professional issues surrounding this data before moving into Chapter 5's detailed breakdown of the actual data collection and preliminary implementation results.

CHAPTER 4: LEGAL, ETHICAL, SOCIAL & PROFESSIONAL ISSUES (LESP)

4.1 Chapter Overview

This chapter evaluates the computing project within the broader context of Legal, Ethical, Social, and Professional (LESP) frameworks. As data science and automated predictive modeling become increasingly integrated into public health and environmental governance, computing professionals must operate under strict regulatory and moral boundaries. This section analyzes compliance with data protection legislation, examines the intellectual property boundaries of third-party environmental data streams, and scrutinizes the ethical implications of using algorithmic imputation to handle missing historical data. Furthermore, it measures the social impact of environmental forecasting systems against established professional codes of conduct, specifically the British Computer Society (BCS) Code of Conduct.

4.2 Legal Issues and Compliance

4.2.1 Data Protection and Privacy Regulations

Although environmental and atmospheric data might initially appear to be purely public and non-personal, the mechanisms of data collection introduce distinct legal liabilities under modern privacy frameworks. This study aligns its data governance with both the Sri Lanka Personal Data Protection Act (PDPA) No. 9 of 2022 and the European Union's General Data Protection Regulation (GDPR), the latter being highly relevant due to the collaborative UK-partnered academic framework of this degree.

The primary legal consideration arises from the spatial positioning of low-cost ground sensors within the PurpleAir network. Unlike government-operated atmospheric stations, many low-cost IoT nodes are purchased, installed, and hosted by private individuals, institutions, or residential households on private balconies and premises. Consequently, the precise geospatial coordinates (latitude and longitude) of a specific sensor node can be cross-referenced with public land registries to identify the physical residence of a citizen. Under both the GDPR and Sri Lankan PDPA, persistent location data that can be linked back to an identifiable natural person constitutes Personal Data. To ensure absolute compliance, this project implements spatial anonymization protocols. Raw coordinate arrays extracted from the PurpleAir API are processed strictly as regional points of interest for Colombo and Kandy, and no attempts are made to collect or store the names, IP addresses, or contact information of individual sensor hosts.

4.2.2 API Terms of Service and Data Ownership

This research depends entirely on data extraction from two major external repositories: the PurpleAir API and the Google Earth Engine (GEE) data catalog containing the European Space Agency's (ESA) Copernicus Sentinel-5P telemetry. Navigating the intellectual property and usage rights of these platforms is essential to ensure legal data acquisition.

- **PurpleAir Data Terms:** The historical extraction of PM_2.5 and meteorological observations utilizes a formalized developer API key system. The platform operates under a tiered consumption model. This project strictly adheres to the fair use boundaries by implementing programmatic data chunking and request throttling, ensuring that the local extraction scripts do not impose an artificial distributed denial-of-service (DDoS) load on the infrastructure. Furthermore, the dataset usage complies with attribution requirements, ensuring that the origin of the ground telemetry is transparently acknowledged.
- **Copernicus Sentinel-5P Data Rights:** The atmospheric NO_2 column density extracted via Google Earth Engine is governed by the European Union's Copernicus data access policy. This legal framework permits free, full, and open access to sentinel data for both academic research and commercial applications, provided that appropriate data citation metrics are maintained. The legal extraction of this data via the GEE API satisfies Google's terms of service, as the computing tasks are processed within cloud sandboxes without circumventing platform security or storage controls.

4.3 Ethical Considerations

4.3.1 Data Integrity, Imputation Transparency, and Synthetic Realism

The most acute ethical challenge within this research lies in data preprocessing, specifically the architectural decision to reconstruct the catastrophic 6-month data blackout within the Kandy ground sensor network using satellite NO_2 proxy variables. Mathematically estimating missing values introduces a profound ethical responsibility regarding scientific integrity.

From an academic perspective, generating "synthetic" or "imputed" ground-level PM_2.5 observations introduces the risk of confirmation bias or data manipulation if not handled with absolute transparency. If the machine learning imputation model over-smooths the historical gaps or fails to represent true variance, subsequent predictive models trained on this data will output highly artificial results. If these results are later cited by public health researchers or local environmental authorities to map respiratory disease trends in Kandy, an inaccurate baseline could lead to misallocated medical resources or flawed urban policies. To mitigate this ethical risk, this project strictly documents and isolates the imputed data phase. In Chapter 5 and all subsequent evaluations, a clear boundary is drawn between physically observed measurements and algorithmically imputed values, ensuring that the performance limitations of the satellite proxy model are candidly exposed.

4.4 Social and Environmental Impact

4.4.1 Public Health Accountability and Misclassification Risks

The deployment of machine learning algorithms for environmental forecasting directly impacts human well-being. Predictive modeling is not a neutral mathematical exercise; it is an active public safety mechanism. For instance, if an advanced LSTM model predicts that the PM_2.5 level in Colombo or Kandy will remain well within the World Health Organization's safe threshold over the next 24 hours, local authorities might refrain from issuing air quality warnings. If the model is incorrect—experiencing a "False Negative" due to overfitting or uncaptured non-linear variables—vulnerable populations, such as children, the elderly, and individuals suffering from chronic asthma, may engage in outdoor activities, directly exposing themselves to dangerous levels of toxic particulates.

Conversely, frequent "False Positives" (predicting severe pollution spikes that do not materialize) can induce unnecessary public anxiety, disrupt daily commerce, and desensitize the population to legitimate future warnings—a psychological phenomenon known as alert fatigue. Therefore, this project treats the minimization of extreme forecasting errors (evaluated through the rigorous optimization of RMSE) not merely as a technical goal, but as a critical social responsibility.

4.5 Professional Frameworks and Codes of Conduct

4.5.1 Compliance with the British Computer Society (BCS) Code of Conduct

As an undergraduate computer science researcher operating within a British-accredited educational framework, the execution of this project must rigorously conform to the professional standards mandated by the British Computer Society (BCS). The design and deployment of the forecasting pipeline directly intersect with three primary pillars of the BCS Code:

- **The Public Interest:** The code dictates that IT professionals must safeguard public health, safety, and the environment. This research is fundamentally built upon this principle, leveraging open-source data science to create a predictive framework aimed at mitigating the invisible health crisis of urban air pollution in Sri Lanka.

- **Professional Competence:** The BCS framework requires professionals to acknowledge their technical limitations and not claim expertise in domains beyond their training. In this project, while the atmospheric chemistry governing PM_2.5 and NO_2 interactions is highly complex, the researcher maintains professional competence by focusing strictly on the *computational, statistical, and algorithmic modeling* of the data, relying on validated empirical formulas from atmospheric science literature rather than making unsubstantiated chemical claims.
- **Integrity:** Computing professionals must maintain absolute honesty in their representation of capabilities and results. This research complies by openly acknowledging the computational constraints of the local hardware, publishing the exact error metrics (including failures and high error variances), and explicitly highlighting the spatial limitations of relying on single low-cost sensor nodes for complex urban topographies.

4.6 Chapter Summary

This chapter successfully mapped the architectural execution of this project against the legal, ethical, social, and professional landscapes. By ensuring strict spatial anonymization to comply with privacy laws, respecting data ownership boundaries, enforcing absolute scientific transparency during the Kandy satellite imputation phase, and operating under the ethical guidance of the BCS Code of Conduct, the project establishes a legally secure and socially responsible foundation. With these systemic guardrails explicitly defined, the report can transition to Chapter 5 to document the concrete physical implementation and preliminary baseline outputs of the machine learning pipeline.

CHAPTER 5: PRELIMINARY IMPLEMENTATION AND DATA ACQUISITION

5.1 Chapter Overview

This chapter documents the practical execution of the first three phases of the system architecture defined in the methodology. It provides a detailed account of the programmatic extraction of multi-modal environmental data, exposing the infrastructural realities and data corruption issues inherent in low-cost sensor networks. The chapter outlines the specific Python and JavaScript pipelines developed to clean, interpolate, and temporally align four years of atmospheric data. Finally, it presents the preliminary benchmarking results of the SARIMAX, Random Forest, and LSTM algorithms, serving as the empirical proof of concept for the proposed multi-modal forecasting framework.

5.2 Ground Sensor Data Extraction and Cleaning

The foundation of the predictive model relies on high-frequency target variables (PM_2.5) and local meteorological features (temperature and humidity). The extraction of this data was executed via a custom Python pipeline interfacing with the PurpleAir RESTful API.

5.2.1 Programmatic Extraction Protocol

Data was extracted for the defined spatial coordinates using Sensor ID 29677 (Colombo) and Sensor ID 12451 (Kandy) for the four-year period spanning January 1, 2022, to December 31, 2025. Because the PurpleAir API limits massive payload requests, a chronological chunking algorithm was developed. The script iteratively requested data on a month-by-month basis, applying a programmed two-second rate-limiting delay between calls to prevent server-side denial of service blocks. The raw UNIX timestamps were systematically converted to Coordinated Universal Time (UTC) and subsequently localized to Sri Lanka Standard Time (UTC +5:30) to ensure accurate alignment with local diurnal cycles.

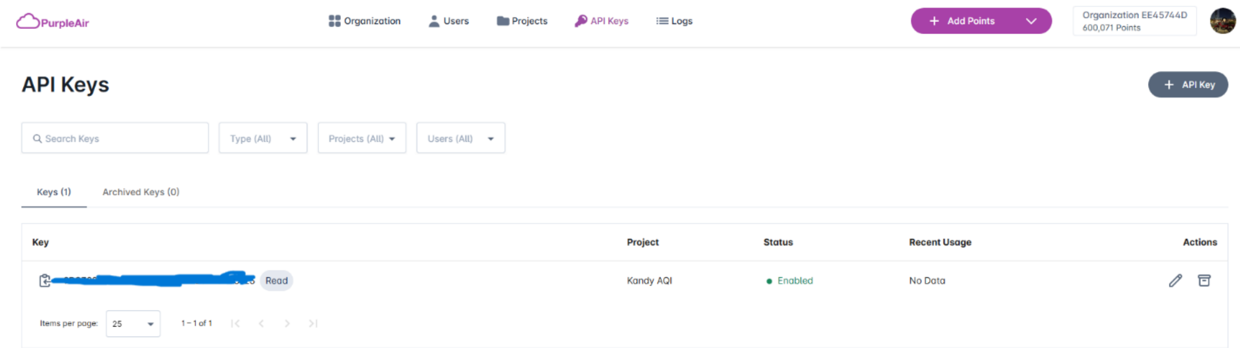

Figure 7 PurpleAir API keys list

Figure 8 PurpleAir API usage dashboard

```python
import requests
import pandas as pd
import time
from datetime import datetime
from dateutil.relativedelta import relativedelta

PURPLE_AIR_API_KEY = "[redacted]"

# two sensors identified for FYP
SENSORS = {
    "Colombo": 29677,
    "Kandy": 12451
}

# 4-Year Timeframe (Jan 1, 2022 to Dec 31, 2025)
START_DATE = datetime( year: 2022,  month: 1,  day: 1)
END_DATE = datetime( year: 2025,  month: 12,  day: 31)

# Fields to extract (Hourly Average = 60 minutes)
FIELDS = "pm2.5_atm,temperature,humidity"
AVERAGE = 60

def fetch_purpleair_data(sensor_name, sensor_index, start_dt, end_dt):  2 usages
    print(f"\n🚀 Starting extraction for {sensor_name} (Sensor ID: {sensor_index})")

    all_data = []
    current_start = start_dt

    # Chunking algorithm: Process month by month to avoid API payload limits
    while current_start < end_dt:
        # Calculate the end of the current month chunk
        current_end = current_start + relativedelta(months=1)
        if current_end > end_dt:
            current_end = end_dt

        # Convert to UNIX timestamps for the API
        start_unix = int(current_start.timestamp())
        end_unix = int(current_end.timestamp())

        print(f"   Fetching data: {current_start.strftime('%Y-%m')}...")

        url = f"https://api.purpleair.com/v1/sensors/{sensor_index}/history/csv"
```

Figure 9 Code for extracting data from API

5.2.2 Gap Analysis and The Colombo Dataset

An initial gap analysis was conducted using the pandas library to evaluate the temporal continuity of the extracted datasets. A perfect four-year hourly dataset should contain exactly 35,064 rows.

For the Colombo sensor, the initial bulk extraction revealed a 90-day missing block. However, this was successfully patched by programmatically merging an earlier three-month Proof of Concept (PoC) dataset collected by the researcher. Following this merge and the removal of overlapping duplicate timestamps, the Colombo dataset yielded 33,668 valid rows. The gap analysis revealed only 37 minor disconnections over the four-year span, with the longest genuine outage lasting just 11.7 days in December 2022.

To prepare the Colombo dataset for Machine Learning ingestion, it was resampled to a strict one-hour chronometric grid. Minor gaps (under 5 hours) were patched using linear interpolation, while the larger 11-day gap was imputed using a historical monthly-hourly profiling algorithm (e.g., calculating the average PM_2.5 for December at 14:00 hours across all four years). This produced a mathematically flawless, 100% complete ground dataset for the Colombo topography.t

5.2.3 The Kandy Blackout Discovery

Conversely, the gap analysis of the Kandy sensor exposed the severe infrastructural vulnerabilities predicted in the literature review. The extraction yielded only 26,283 rows (approximately 75% data completeness). The analysis script detected 324 separate network disconnections. Most critically, it isolated a catastrophic 201-day hardware blackout spanning from August 29, 2024, to March 18, 2025.

Because Deep Learning algorithms (LSTMs) and seasonal statistical models (SARIMAX) strictly require an unbroken sequence of time, standard linear interpolation or historical averaging cannot be applied to a six-month void without severely corrupting the model's predictive integrity. The discovery of this massive failure directly triggered Phase 4 of the system architecture: the necessity for Satellite Imputation.

5.3 Satellite Data Acquisition via Google Earth Engine

To introduce macro-level atmospheric context and provide the necessary proxy variables for the Kandy imputation, a secondary extraction pipeline was executed to acquire Tropospheric Nitrogen Dioxide (NO_2) column density.

5.3.1 Sentinel-5P Extraction

The extraction was executed within the Google Earth Engine (GEE) environment using a custom JavaScript algorithm. The script queried the COPERNICUS/S5P/OFFL/L3_NO2 image collection. To ensure exact spatial correlation, the algorithm utilized an ee.Reducer.mean() function, filtering the global satellite imagery strictly to the geospatial points of the Colombo and Kandy ground sensors at a spatial scale of 1,113.2 meters.

5.3.2 Handling Cloud Masking

Optical satellite sensors cannot penetrate dense cumulonimbus clouds, which are highly prevalent during Sri Lanka's Southwest and Northeast monsoons. Consequently, the GEE extraction algorithm included a .filter(ee.Filter.notNull()) protocol to automatically drop days where the satellite's view of the troposphere was completely obscured. The resulting dataset successfully captured the valid daily NO_2 telemetry for the four-year period, exported via the Export.table.toDrive function as a localized CSV file, ready for temporal fusion with the ground sensor matrices.

Sentinel5P_Colombo_NO2_Clean.csv

	A	B	C	D
1	system:index	Date	NO2_Density	.geo
2	20230101T074405_	2023-01-01	5.53E-05	{"type":"MultiPoint","
3	20230103T084905_	2023-01-03	8.66E-05	{"type":"MultiPoint","
4	20230104T082905_	2023-01-04	6.67E-05	{"type":"MultiPoint","
5	20230109T083405_	2023-01-09	6.18E-05	{"type":"MultiPoint","
6	20230110T081905_	2023-01-10	8.33E-05	{"type":"MultiPoint","
7	20230111T075905_	2023-01-11	1.14E-04	{"type":"MultiPoint","
8	20230112T073905_	2023-01-12	5.61E-05	{"type":"MultiPoint","
9	20230113T071905_	2023-01-13	1.91E-04	{"type":"MultiPoint","
10	20230114T084405_	2023-01-14	1.05E-04	{"type":"MultiPoint","
11	20230115T082405_	2023-01-15	1.26E-04	{"type":"MultiPoint","
12	20230119T084806_	2023-01-19	2.99E-05	{"type":"MultiPoint","
13	20230120T082806_	2023-01-20	8.72E-05	{"type":"MultiPoint","
14	20230125T083806_	2023-01-25	5.49E-05	{"type":"MultiPoint","
15	20230126T081806_	2023-01-26	1.16E-04	{"type":"MultiPoint","
16	20230127T075806_	2023-01-27	9.82E-05	{"type":"MultiPoint","
17	20230128T073806_	2023-01-28	3.00E-05	{"type":"MultiPoint","
18	20230205T082806_	2023-02-05	2.14E-05	{"type":"MultiPoint","
19	20230208T073306_	2023-02-08	2.95E-05	{"type":"MultiPoint","
20	20230210T083806_	2023-02-10	5.21E-05	{"type":"MultiPoint","
21	20230211T081806_	2023-02-11	4.86E-05	{"type":"MultiPoint","
22	20230212T075806_	2023-02-12	1.29E-04	{"type":"MultiPoint","
23	20230213T073806_	2023-02-13	1.20E-04	{"type":"MultiPoint","
24	20230214T072306_	2023-02-14	1.11E-04	{"type":"MultiPoint","
25	20230215T084306_	2023-02-15	4.48E-05	{"type":"MultiPoint","
26	20230216T082306_	2023-02-16	1.47E-04	{"type":"MultiPoint","
27	20230217T080306_	2023-02-17	9.80E-05	{"type":"MultiPoint","
28	20230218T074806_	2023-02-18	8.43E-05	{"type":"MultiPoint","
29	20230221T082806_	2023-02-21	4.07E-05	{"type":"MultiPoint","
30	20230222T081306_	2023-02-22	1.85E-04	{"type":"MultiPoint","
31	20230223T075306_	2023-02-23	7.07E-05	{"type":"MultiPoint","
32	20230226T083806_	2023-02-26	8.36E-05	{"type":"MultiPoint","
33	20230227T081806_	2023-02-27	1.09E-04	{"type":"MultiPoint","
34	20230305T080306_	2023-03-05	4.19E-05	{"type":"MultiPoint","
35	20230306T074806_	2023-03-06	4.37E-05	{"type":"MultiPoint","
36	20230307T072806_	2023-03-07	1.39E-04	{"type":"MultiPoint","
37	20230308T084806_	2023-03-08	9.27E-05	{"type":"MultiPoint","
38	20230309T082806_	2023-03-09	1.39E-04	{"type":"MultiPoint","
39	20230311T075306_	2023-03-11	1.84E-04	{"type":"MultiPoint","
40	20230312T073306_	2023-03-12	4.14E-05	{"type":"MultiPoint","
41	20230313T085306	2023-03-13	9.64E-05	{"type":"MultiPoint","

Figure 10 Sentinel5P extracted data

5.4 Feature Engineering and Temporal Alignment

The core technical challenge in creating a multi-modal forecasting framework lies in the mathematical reconciliation of data streams operating on radically different temporal frequencies. While the PurpleAir ground sensor array generates continuous, high-frequency hourly observations ($f = 1$ Hour), the Copernicus Sentinel-5P satellite collects macro-level atmospheric column data on a low-frequency, daily interval ($f = 1$ Day). To build a cohesive training matrix for machine learning models, a highly robust feature engineering and temporal alignment pipeline was developed in Python using the pandas and numpy libraries.

5.4.1 Left-Join Broadcast Protocol

To resolve the temporal asymmetry, a specialized chronological broadcasting engine was constructed. The hourly datetime index of the cleaned ground sensor matrix (colombo_ml_ready.csv) was normalized, isolating the pure calendar date string (e.g., YYYY-MM-DD) into a temporary indexing feature column. The daily satellite dataset (Colombo_NO2_4Years_2022_2025.csv) used this calendar date as its primary key. A relational Left-Join merge protocol was then executed, anchoring the operation to the high-frequency ground dataset. This operation successfully mapped and duplicated the single daily tropospheric NO_2 column density observation across all corresponding 24 hourly rows of that specific day. This architectural choice ensures that the macro-level pollution background of the urban troposphere acts as a continuous ambient feature for the micro-level ground predictions.

5.4.2 Cross-Scale Satellite Interpolation

As discovered during data acquisition, atmospheric cloud cover obscured the satellite's optical sensors on approximately 60% of the days across the 2022–2025 timeline, resulting in 22,056 missing hourly blocks upon completion of the left join. Because trace gases like NO_2 exhibit strong temporal correlation and maintain relatively stable macro-baselines day-to-day (barring sudden meteorological changes), a dual-directional linear interpolation routine was compiled. By applying the .interpolate(method='linear', limit_direction='both') function, the algorithm smoothly bridged the temporal voids by calculating a steady gradient between the closest valid historical and future satellite captures. This technique brought the final missing NO_2 count to zero, completing an unbroken 35,064-row spatial-temporal fused training matrix (colombo_final_fused_4years.csv).

5.4.3 Multi-Horizon Feature Generation

To transform the static feature matrix into a dynamically responsive time-series structure capable of training tree-based and sequential neural network architectures, multi-horizon feature engineering was executed. Three distinct classes of features were generated:

1. **Temporal Cyclical Components:** To allow non-linear models to interpret human behavioral cycles (such as morning and evening traffic rush hours or weekend commercial spikes), the datetime index was deconstructed into discrete integer columns representing Hour (0–23), DayOfWeek (0–6), and Month (1–12).

2. **Autoregressive Lag Variables:** To capture immediate historical dependencies, the target variable (PM_2.5) was programmatically shifted backward in time. Three lag horizons were constructed: a short-term 1-hour lag (Lag_1H), a 2-hour lag (Lag_2H), and a seasonal 24-hour lag (Lag_24H). These parameters serve as an explicit memory mechanism for the non-sequential tree-based models.

3. **Statistical Rolling Windows:** A 24-hour rolling mean calculation window (Rolling_Mean_24H) was engineered by executing a shifted rolling function. This variable calculates the average pollution level over the prior day, providing the models with a steady indicator of macro-trends and smoothing out localized, transient noise.

Rows containing NaN values generated by the initial lagging operations (the first 24 hours of 2022) were stripped from the matrix, yielding a final, clean, multi-modal feature array ready for algorithmic benchmarking.

5.4.4 Primary Research Findings and Needs Analysis

MS Forms was used for collecting survey data and within the time-period of 4 months, 52 responses from respondents across Sri Lanka was recorded. Then the recorded survey data was exported to an excel sheet which was transformed into informative graphs using Power-Bi software.

Public Awareness & Requirements for Air Quality Forecasting in Sri Lanka

As part of my Final Year Project (FYP), I am developing an AI-powered "Virtual Sensor" system and mobile dashboard to predict air pollution levels 24 hours in advance for Colombo and Kandy. Your feedback will directly shape the features and design of this early warning app.. All responses are completely anonymous and will be used strictly for academic research purposes. Thank you for contributing to this project!

1. Which city do you primarily live or work in? *

- Colombo (Coastal/Commercial)
- Kandy (Valley/Hill Country)
- Other

2. Do you or anyone in your direct household suffer from respiratory conditions (e.g., Asthma, wheezing, allergies)? *

- Yes
- No

3. How would you describe the air quality in your area currently? *

- Very Clean
- Moderate
- Polluted (Dusty/Smoky)

4. Do you currently check air quality levels (AQI) before leaving home? *

- Yes, using an app/website (e.g., Weather App, IQAir)
- Yes, using the US Embassy Monitor
- No, I just look at the sky/visibility
- No, I don't check at all

5. Do you feel the current air quality information covers your specific neighborhood (e.g., your exact street/suburb)? *

- Yes, it is accurate for my exact area.
- No, it only shows the common areas.
- I am not sure.

6. Have you ever been surprised by a sudden bad air day (e.g., unexpected smog or dust) that you were not prepared for? *

- Yes, often.
- Sometimes.
- Never.

Figure 11 Survey questions 1

7. If you knew 24 hours in advance that air quality would be "Hazardous" tomorrow, what actions would you realistically take? (Select all that apply) *

- [ ] Wear a KN95/PM2.5 mask outdoors
- [ ] Keep children/elderly indoors
- [ ] Keep home/car windows closed
- [ ] Cancel or reschedule outdoor sports/workouts
- [ ] I wouldn't change my routine

8. How would you prefer to receive this 24-hour early warning? *

- ○ Immediate Push Notification (Pop-up on phone)
- ○ A morning summary notification (e.g., at 7:00 AM)
- ○ SMS / Text Message
- ○ No notifications, I will just open the app to check

9. What information is the most important to see when you first open the app? (Choose Top 2) *

- Tomorrow's 24-hour forecast graph
- A simple color-coded dial (Green/Yellow/Red)
- A map showing the pollution cloud over my city
- Specific health advice (e.g., "Safe for running today")

10. If the app explains *why* the air is bad (e.g., "High pollution today is due to heavy traffic" vs. "Due to Indian Monsoon Smog"), would that help you trust the app more? *

- ○ Yes, definitely.
- ○ Maybe, but I only care about the final number.
- ○ No, I don't need to know the cause.

11. What would be your preferred language for receiving these health warnings? *

- ○ English
- ○ Sinhala
- ○ Tamil

Figure 12 Survey questions 2

Primary Residence Respondents are mostly from Colombo and Kandy (~20 each), with smaller numbers from Gampaha, Kalutara, and scattered representation from other areas.

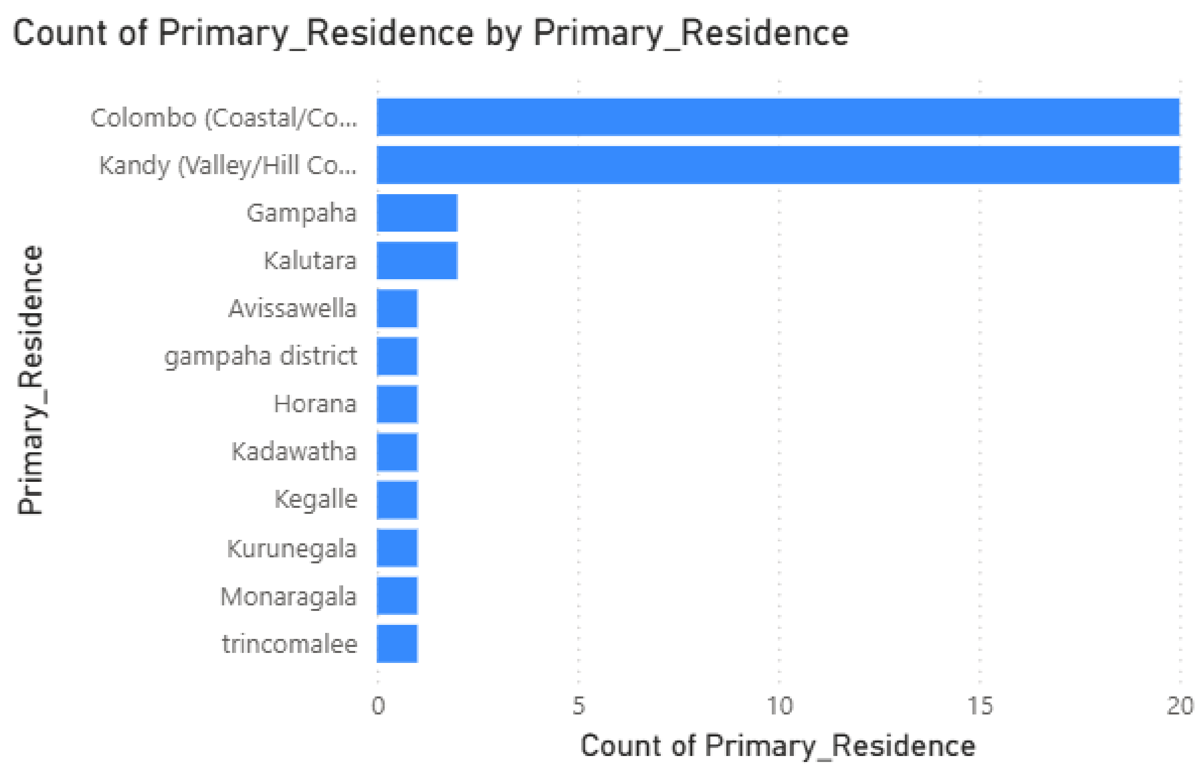

Figure 13 primary residence graph

Respiratory Issues 63% of respondents have no respiratory issues, while 37% do.

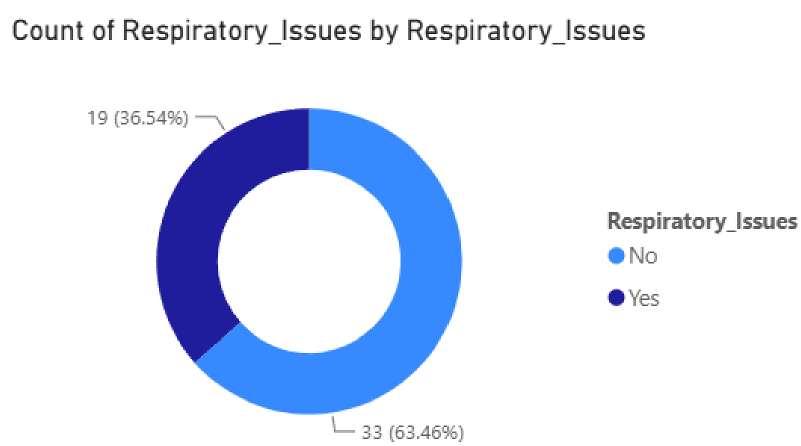

Figure 14 Respiratory issues graph

Current AQ Perception The majority (~30) perceive current air quality as "Moderate." About 11 each rated it as "Polluted (Dusty/Smoky)" or "Very Clean."

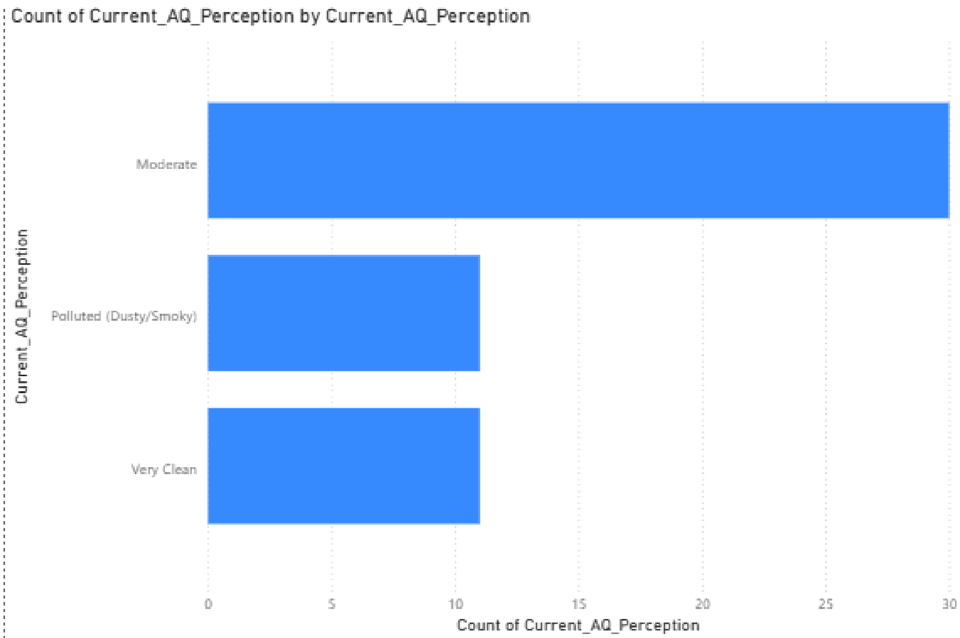

Figure 15 Current AQ Perception graph

Checking AQI 62% don't check AQI at all. 31% just look at the sky/visibility. Only 8% use an app or website.

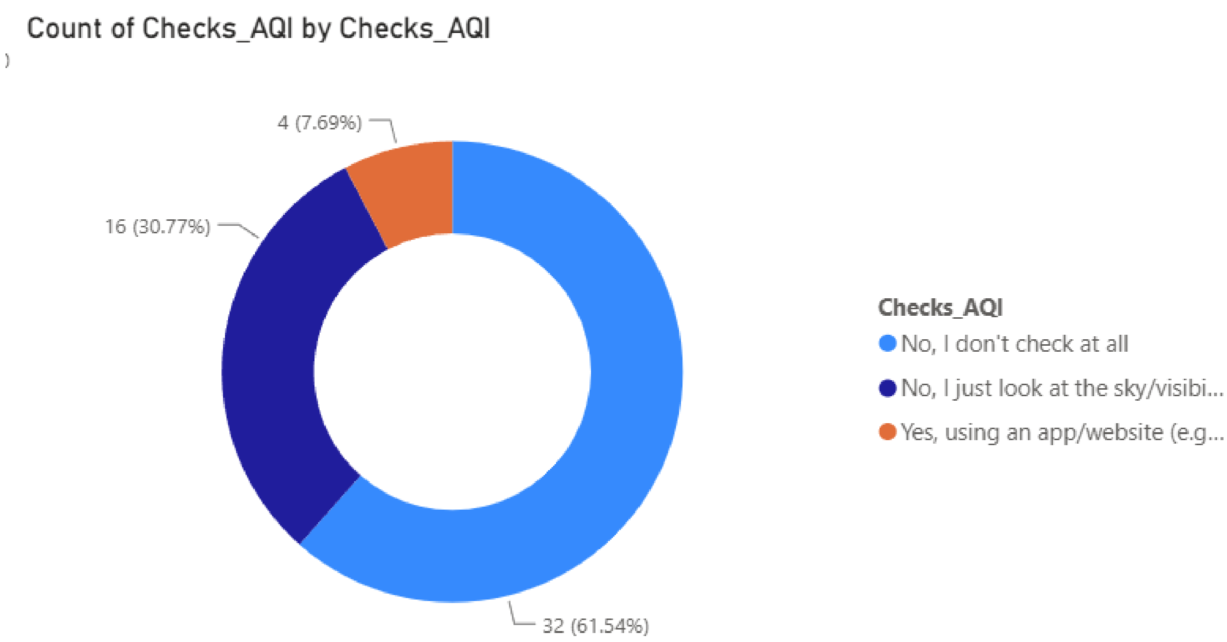

Figure 16 Checking AQI graph

AQ Coverage Awareness Nearly half (48%) are unsure if air quality data covers their exact area. 40% believe it only covers common areas, and just 12% think it's accurate for their specific location.

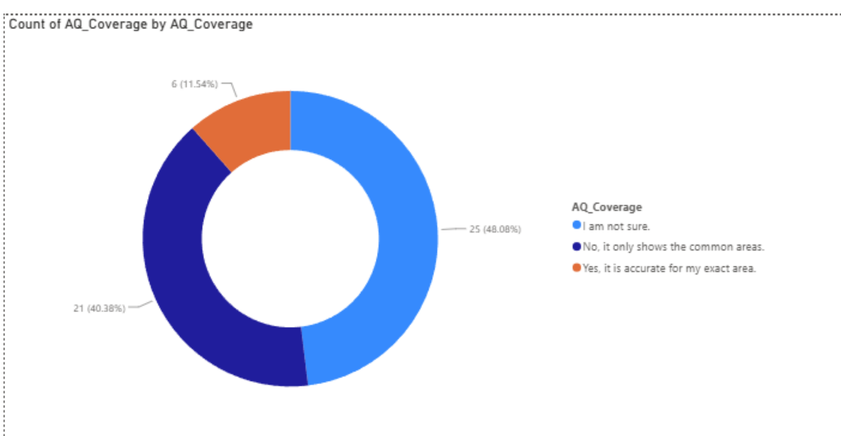

Figure 17 AQ Coverage Awareness graph

Unexpected Bad Air Days 71% say they "sometimes" experience unexpectedly bad air days, 15% say "yes, often," and only 13% say "never."

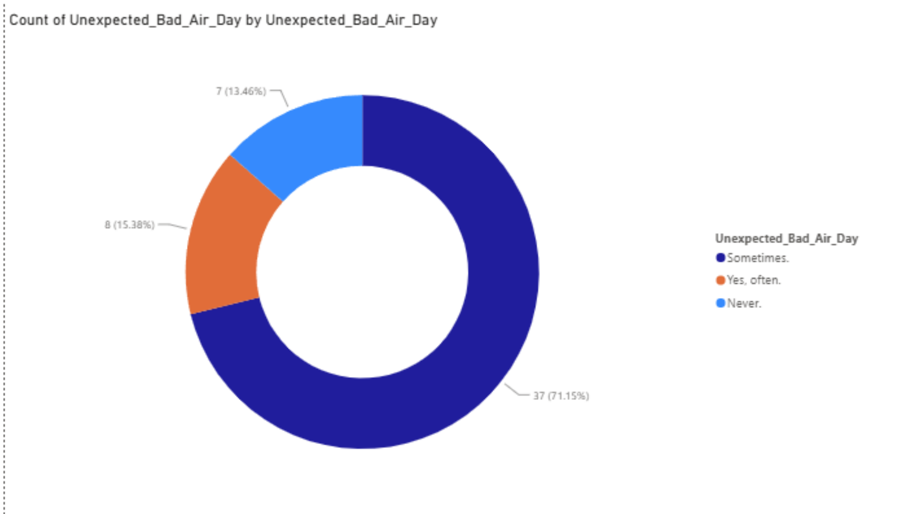

Figure 18 Unexpected Bad Air Days graph

Actions Taken When air quality is bad, the most common response is wearing a KN95/PM2.5 mask (~32 people), followed by keeping windows closed (~25), canceling outdoor workouts (~23), keeping children/elderly indoors (~19), and only ~9 said they wouldn't change their routine.

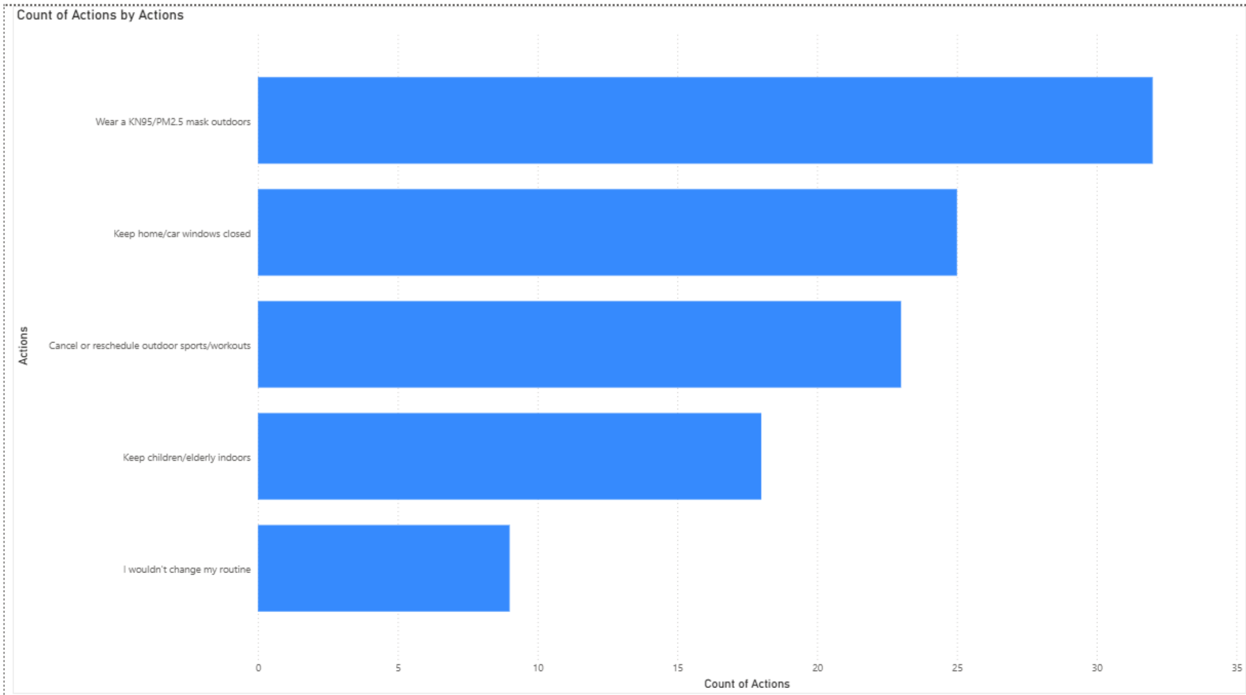

Figure 19 Actions Taken graph

Warning Delivery Method Push notifications are most preferred (~21), followed by SMS/text (~17), morning summary notifications (~10), and only ~3 prefer opening the app themselves.

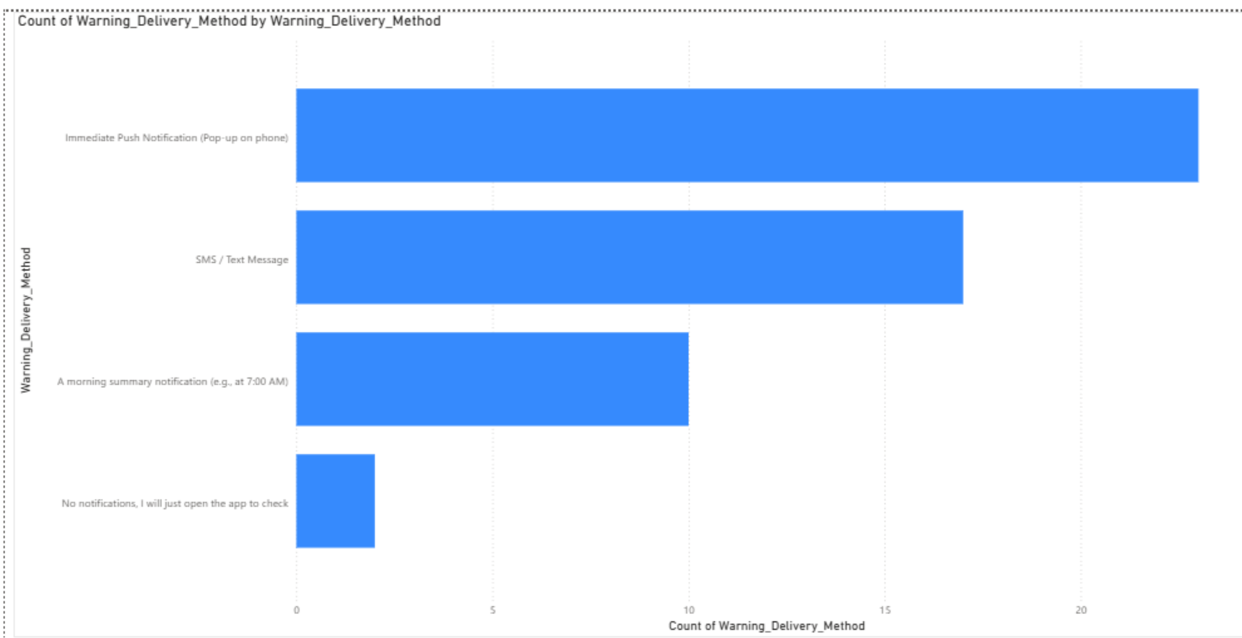

Figure 20 Warning Delivery Method graph

Preferred Display Format Respondents ranked a simple color-coded dial (Green/Yellow/Red) as their top choice. A 24-hour forecast graph came second, followed by a pollution cloud map, then specific health advice.

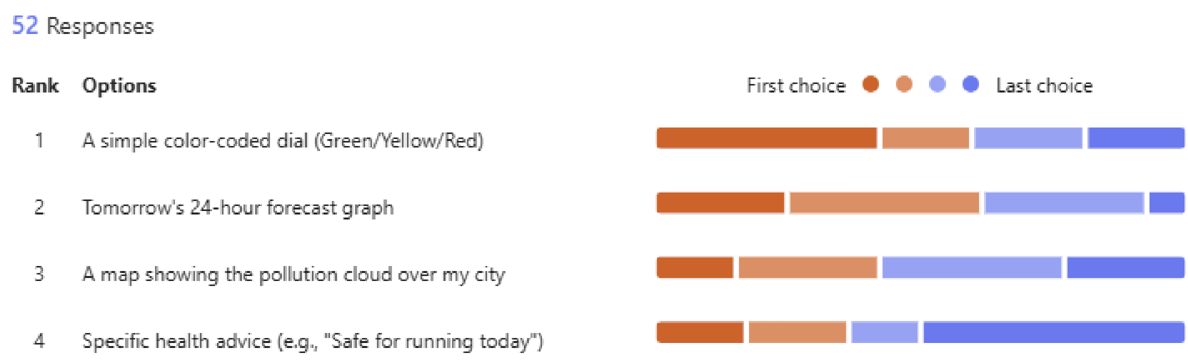

Figure 21 Preferred Display Format graph

Preferred Language English is overwhelmingly preferred (~47 respondents). Sinhala had ~4 and Tamil ~2.

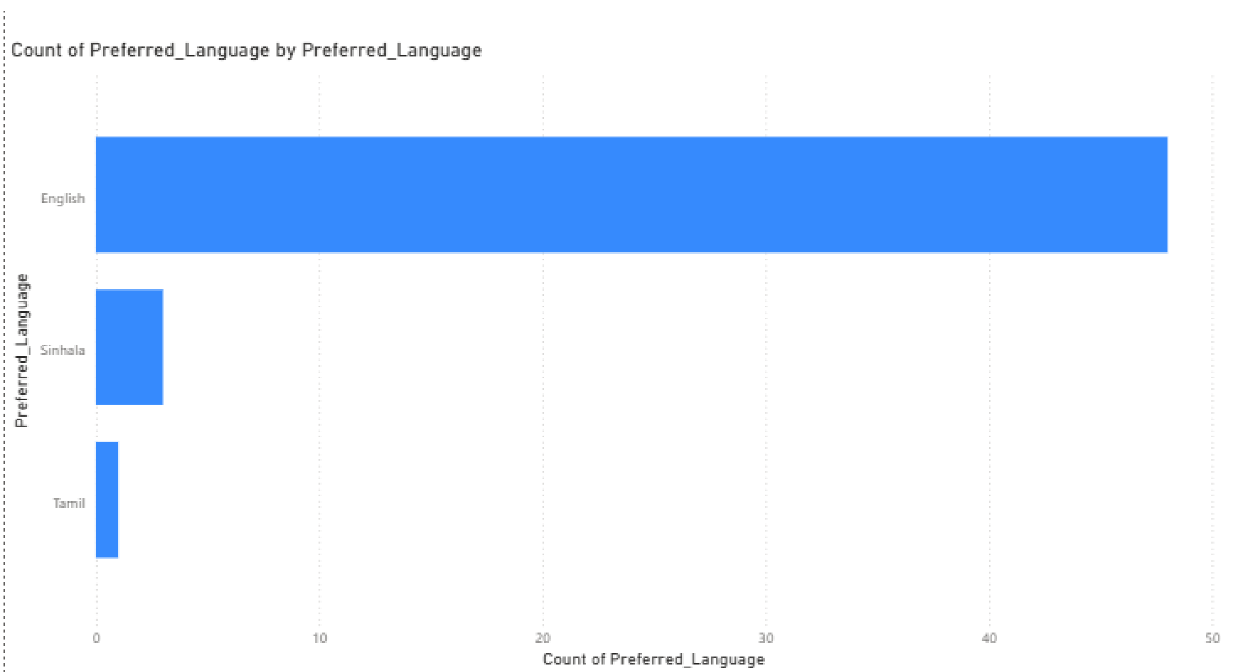

Figure 22 Preferred Language graph

Trust & Explanation 77% feel that explaining the *cause* of air quality definitely increases their trust in the data. 21% only care about the final number, and just 2% don't need to know the cause at all.

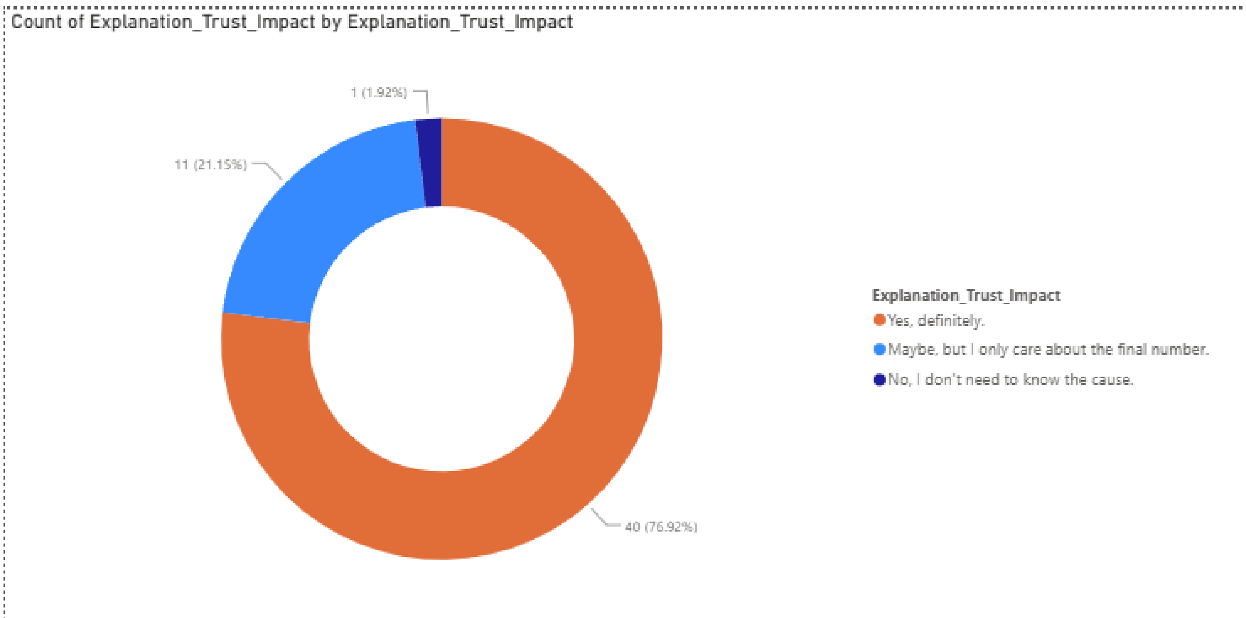

Figure 23 Trust & Explanation graph

Detailed Statistical Breakdown of Public Survey Metrics

To translate public sentiment into concrete software specifications, the empirical data gathered from the survey (N=52) is categorized across three major analytical dimensions: health vulnerability, information access barriers, and interface/functional preferences.

1. Baseline Cohort Demographics and Health Profiles

The geographic distribution of respondents shows a balanced representation within the target urban areas under investigation, with approximately **38.5%** (n=20) originating from Colombo and **38.5%** (n=20) from Kandy, while the remaining **23%** are distributed across adjacent high-density zones like Gampaha and Kalutara. This localized distribution ensures the data accurately reflects the primary populations affected by urban pollution corridors in Sri Lanka.

Cross-referencing this distribution with household health profiles reveals a major systemic vulnerability: **36.5%** (n=19) of respondents explicitly confirm that a member of their immediate household suffers from a chronic respiratory illness (such as asthma, wheezing, or airborne allergies). This significant percentage confirms that more than one-third of the target user base has a critical physical dependency on accurate environmental alerts, elevating the software platform from a convenience tool to an essential healthcare utility.

2. The Quantification of Information Gaps and Spatial Uncertainty

The survey data highlights a critical failure in current public communication frameworks regarding environmental metrics. While the absolute majority of respondents (**57.7%**, n=30) subjectively categorize their local air quality as "Moderate," their capacity to make data-backed health decisions is severely restricted:

- **Infrastructural Blind Spots: 48.1%** (n=25) of the population are entirely uncertain if current environmental information covers their specific neighborhood or street. An additional **40.4%** (n=21) believe coverage is strictly confined to major centralized nodes. Only a marginal **11.5%** (n=6) feel confident that current data represents their exact spatial coordinates.

- **The Volatility Factor:** This spatial blindness is heavily penalized by rapid environmental changes, with **71.2%** (n=37) stating they "sometimes" face unexpectedly hazardous air days unprepared, and **15.4%** (n=8) experiencing this volatility "frequently."
- **Systemic Non-Engagement:** Consequently, **61.5%** (n=32) of respondents confess they never check the Air Quality Index (AQI) before leaving home. Instead, **30.8%** (n=16) resort to crude visual heuristics (scanning the horizon or checking visibility), while a trivial **7.7%** (n=4) actively utilize a modern web or mobile software tool.

This establishes a clear mandate for an automated platform that proactively bridges the spatial tracking gap without demanding active technical searching from the user.

3. Deriving Functional and Non-Functional System Requirements

The survey data serves as the definitive source for mapping user expectations directly onto software architectural layers, as broken down below:

Target System Attribute	**Empirical User Preference**	**Technical Software Specification**
Primary UI Visualization	1. Color-Coded Gauge Dial (High) 2. 24-Hour Forecast Graph (Medium) 3. Spatial Heatmaps / Health Advice (Low)	Implement a centralized, dynamic SVG/Canvas color-wheel gauge (Green/Yellow/Red) linked to an interactive chart rendering 24 hourly steps via a predictive regression backend.

Target System Attribute	Empirical User Preference	Technical Software Specification
Notification Pipeline	1. Push Alerts (n=21) 2. SMS Text Messages (n=17) 3. Summary Digests (n=10) 4. Manual Fetching (n=3)	Design an active notification daemon using Firebase Cloud Messaging (FCM) or Twilio API to fire background payload pushes whenever forecasted AQI crosses defined safety thresholds.
Language Localization	English: **90.4%** (n=47) Sinhala: **7.7%** (n=4) Tamil: **3.8%** (n=2)	Set English as the primary compilation language for the user interface, while establishing localized resource files to support multilingual scaling in subsequent builds.

Target System Attribute	Empirical User Preference	Technical Software Specification
Algorithmic Transparency	**76.9%** (n=40) state that root-cause explanation definitely increases data trust.	Integrate an automated **Explainable AI (XAI)** translation engine on the backend that reads SHAP feature importance vectors and outputs natural language logs.

Table 3 System Requirements table

Primary Research Interviewees

1. Mr. Janindu Chandrasena from Peradeniya University who is a Research Assistant | Air Quality Analyst

He's currently working on a project (Respire2) where they measure the air quality all around Sri Lanka from BlueSky sensors and they also get data from patients from local hospitals evaluating the respiratory symptoms, what they want to find is whether there is a correlation between air quality and the symptoms among the people. Apart from that they focus on the air quality in Kandy area since they are students at Peradeniya University as well as Kandy city being highly populated and polluted. (they mainly measure PM 2.5 particles). He emphasized that Kandy being situated in a basin contributes heavily to the pollution (air comes into the Kandy city and cannot escape because of the valley effect) apart from this the roads are not of perfect shape and corruption among officials and well as drivers also contribute to heavy vehicle smoke. Therefore even a more densely populated city like Colombo has better Air Quality because of the geographical differences as well as better traffic planning. When it comes to the biggest headaches when either deploying or maintaining the sensors is being unable to afford state of the art sensors (they use low cost sensors given to them by a research grant) as well as them only able to maintain by a person with knowledge to the subject and the chemistry behind it. They sometimes also lack the permission from government to place these sensors in the most efficient places namely around the Kandy lake as the sensor deployed in Torrington park is not enough to cover whole of Kandy. He emphasized that even though the BlueSky sensor is low cost, the data collected from it is reliable as it's renowned around the world. Although no official prediction models are made public by the CEA, they sometimes do warn of bad air quality days, they have established data points across Jaffna, Batticaloa and Colombo etc. Mr. Janindu and his team has also established a vast network of sensors spanning 24 districts but sometimes the sensors breakdown and it's difficult to repair/replace them as they can be in remote areas. They have used satellite data for past research analysis (specifically heat images), but not for air quality predictions. If they notice a spike in Air Quality they rely on past data to pinpoint what might be the reason behind the spike (e.g 5-8 a.m during weekdays as traffic is high during those hours). When I inquired him about whether the government or fellow researchers might be open to use predictions made using A.I he replied that even A.I based models need high quality and quantity of past data to train well and it also heavily depends on the people in power. If a well trained A.I gives the same or very close predictions to real sensors it can be considered although some people might still be skeptical about how the results or the science behind the predictions is acquired. Looking at things scientifically there needs to be a reason for whatever conclusion that is met. This is important for the public to accept the predictions as well. He emphasized that an explainable A.I will do good for the acceptance of results. If a 24 hour prediction dashboard were to exist that gave the government much needed time to act for a future event it would certainly help but a noticeable change must be done by the government as well as the public for it to take effect. For example, reducing traffic greatly reduces emission which in turn improves the air quality, but Sri Lankan public transport system leaves a lot to be desired which will make people opt for personal transport. He further elaborated that while the current air quality is not the best its still far from situations such as having to close schools to protect the children etc. This does not count for unforeseen circumstances such as a large-scale burn that might happen by an emergency. A lot of workplaces have drills and protocol to follow in case of such situations and there are also policies for employees which protect them from harm such as a level of Air Quality that is considered optimum for working environment. There are also limits for industries which can release such pollutants to the environment.

To conclude the interview he spoke out that there is no real-time system yet to protect sensitive individuals such as traffic police, children who are outside as soon as school is over (high-traffic time) that are constantly being exposed to air pollution, so my project could help such people that could alter their shifts and give medical outlooks.

Figure 24 Primary research interview pic 1

2. Mr. K.H.T.R Kumara from Central Environmental Authority (Senior Air Quality Specilaist)

Having more than 6 years of experience (1 year experience collecting data from the Kandy control center and performing analysis before sending it to the Battaramulla headquarters, was involved in many projects which include making the old road, one-way, unfortunately it did not change the pollution levels), he mentioned that while the expensive BAM machines are only deployed in Battaramulla and Kandy (There is also a mobile BAM machine in Colombo), other places like Anuradhapura, Kurunegala and Hambantota etc. just use ground level sensors. The main issue when it comes to maintaining the expensive machinery is budgeting as these sensors cost around 5 million LKR for annual maintenance as well as a Rs.100,000 monthly servicing as well. The main difference between BAM machines and sensors is that the former scans fine particulate matter for a more accurate reading while the much simpler sensors are easier to maintain and calibrate but isn't as accurate as the expensive machinery. Mr. Kumara also cites the main issue for Kandy is the valley trapping effect and for Colombo the main issue being trans-boundary pollution (for coastal areas, when it reaches inside the country it's diluted. The CEA does not have island wide-coverage as of now but there are plenty of NGO's that aid with their own sensors. The data is published on their website and updated every 15 minutes, Mr. Kumara also mentioned that Air Quality is deeply connected to climate and human deeds, therefore predicting it for longer times is improbable and its better to predict for a week window instead, therefore the CEA is not currently using any predictive tools, but he did mention that there are similarities if we take a look at a graph from last year and compare it to the current year same time frame. He stressed that the current laws they cannot do anything to combat the pollutants from India as it's very hard to prove it carried over from there, the best they can do is alert sensitive groups using media, currently no satellite imagery is being used to monitor Air Quality. Mr. Kumara mentioned that the SHAP explanation is a good thing to implement as black box predictions are tough to trust, but it's a difficult endeavor, If I were able to provide a dashboard that was able to reliably predict the air quality, future policies can help the CEA take action such as public transport overhauls etc.

Figure 25 Primary research interview pic 2

3. Mr. Duminda Herath from Duminda Auto Electronics (HYBRID ,EV, EFI Tuning ,Scan & Repair Center)

More than 20 years of experience in the motor services industry including 5 years in Japan he is an expert in EFI Turning which is relevant to the project as it is focused on testing vehicle emissions. I inquired him about the accuracy of emission tests and he said they are almost 100% correct, he cemented his answer by saying that many people try to get a different reading for their vehicle by going to another emission testing place but it does not change the result till the vehicle is fixed. There are two main emission testing companies in Sri Lanka which is DriveGreen and Laughs, both of which use mostly the same testing procedures. He stressed that in newer vehicles (efi vehicles) it's almost impossible to fake the sensor data to dupe emission readings. On older vehicles without electronic sensors, it is possible to dupe the readings by cutting off diesel valves etc. but they can be found by random spot checks by the police. By this we can confirm that the NO_2 emission readings in Sri Lanka are highly accurate. Moving onto the emission testing machines itself he said the main challenge is the price of buying them but they are not that expensive to maintain. There is no difference in the failure rates of vehicles between Colombo and Kandy, it has more to do with newer vehicles failing less than a regional issue. He mentioned that while salty air from the coastal places might degrade other parts of a vehicle, it does not affect the exhaust system. The seasonal changes (colder months) within Kandy also do not change the emission readings as newer vehicles always get to an ideal temperature within a minute as they are designed to function in many countries which have different average temperatures. He implied that regular checks on specifically older vehicles can even further reduce the poisonous gases released into the environment but it's not practical to reduce those emissions to 0. Currently there are no prediction models used in the emission testing sector. When a reading is taken on a vehicle, outside pollution will not affect the reading in any way, he cemented this by replying that for an emission test to get outside interference the pollution levels must get to a point where humans won't be able to survive in such an environment. When I took insights about my prediction model he insisted that within a 24 hour window the air quality might not change much but if I extended the time-frame to about 6 months I could find some varying readings, this was a contradictory point to my other interviewees who insisted that air quality can have high spikes during rush hours etc. However when I inquired him further about his answer he said that he was talking about a district rather than a specific area, this answer made much more sense as even with an air quality spike, within a few minutes it will disperse and it will not show up on a district wise level. He further stressed that inhaling 0.4% CO concentration in the air we breathe is enough for a fatal outcome, but regarding emission tests, the pass rate for a vehicle is at 3% and that's because the 3% will rapidly mix with the outside air will ventilate. From 100 years ago till now with all the new fuel burning machines, the environmental NO2 levels have only increased to the 4^{th} decimal place, but even that small increase can still affect the human body. He mentioned that there can be headaches while working on older vehicles, as a suggestion he urges the government to push electric vehicles to the public by using trade in offers where people can exchange their old cars for newer ones but destroying the old one so that the count of vehicles in the country stays the same. The passing benchmark for the emission tests themselves have also gotten tougher with the CO rate dropping to 3% from 4.5% in the past. An explainable A.I could help the government take action if the sudden spikes were related to vehicle emissions by implementing policies as he mentioned above.

He said that he himself has air quality sensors in his home and they record many spikes during different times of day because of ventilation levels etc. As for the concluding remarks he mentioned that in regards to vehicle pollution it is heavily controlled but other industries like factories as well as domestic burning affects the environment negatively in a higher impact.

Figure 26 Primary research interview pic 3

4.Ms. M.G.J.S Karunathilaka from Maam Biscuits Lanka (Pvt) Ltd who is the Quality Assurance Executive

This interviewee was picked for looking at the industrial side of things, Currently working as the Quality Assurance Executive at Maam Biscuits Lanka, She mentioned that they are utilizing heavy machinery such as boilers, large-scale ovens and generators and the primary fuel source for them is electricity. 1 – 3 pm are their peak hours for logistics operations but the trucks are brought to the facility early in the morning for loading, As per my observations inside the factory the room was not 100% sealed therefore weather effects can be seen. Currently there are no air quality sensors deployed within the facility, only observed visually through the naked eye. No predictive modeling tools are utilized currently. She further mentioned that there are no issues to the people working inside the factory from adverse air quality but emissions could be released to the outside. Baking is done 1-2 days (entire day) per week and the rest of the days are used for packing the biscuits. Baking days produce the most emissions. When I inquired her about whether they would be open to trusting A.I explanations and predictions regarding Air Quality, she was positive to the idea and mentioned with the current technological advances notably A.I, the company needs to adapt to new technologies without lagging. Elaborating on that she also mentioned that the SHAP analysis system will help with trusting the system more as there might be hesitation to trust the predictions without it. At the end she mentioned that the specific reason as to why the air might be bad is not very relevant and they will take the same precautions regardless of the reason.

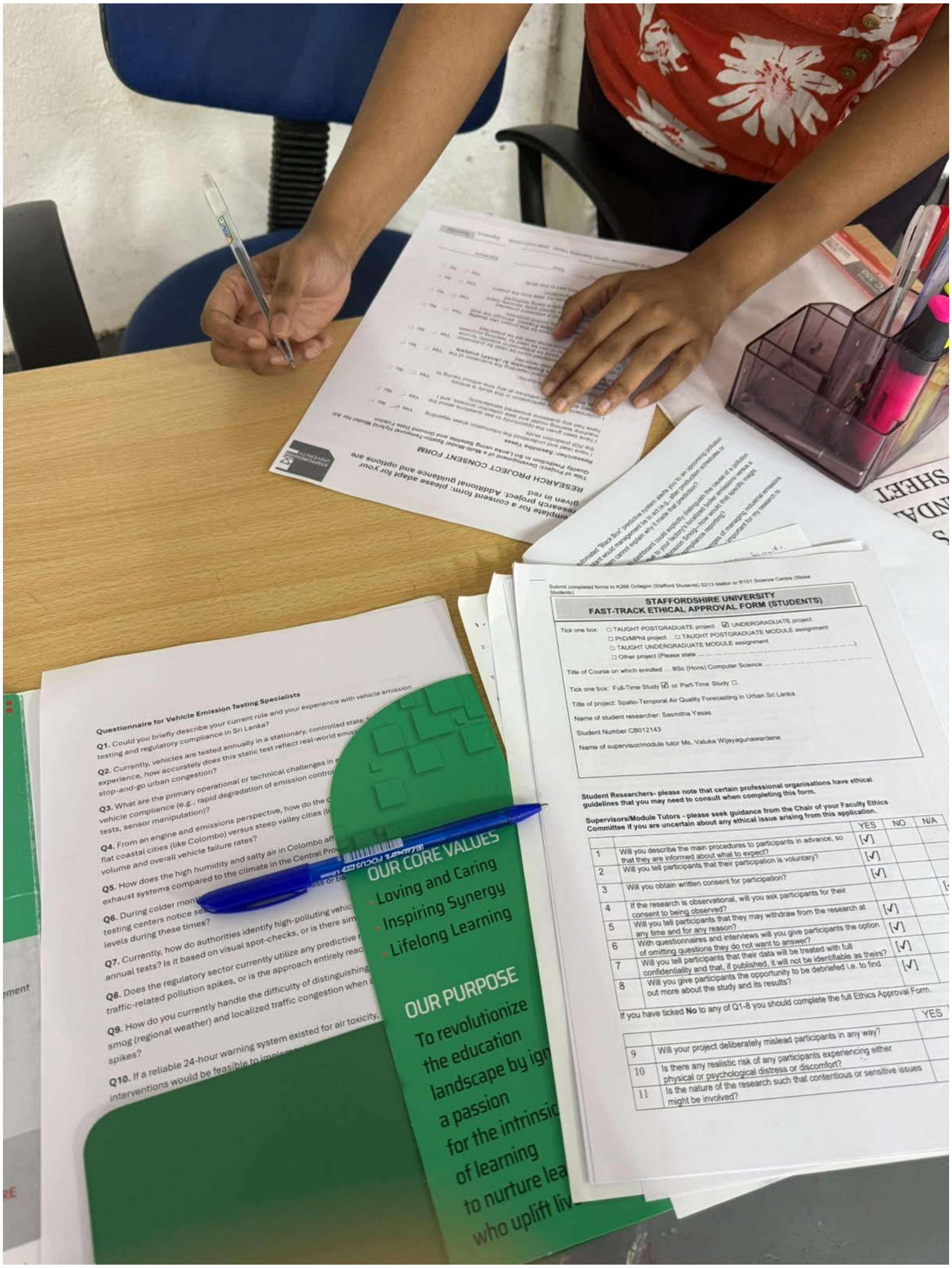

Figure 27 Primary research interview pic 4

Topographical Validation: The "Tale of Two Cities"

The selection of Colombo and Kandy for comparative analysis was explicitly validated by both academic and regulatory experts. Mr. Janindu Chandrasena (Air Quality Analyst, Peradeniya University) and Mr. K.H.T.R. Kumara (Senior Air Quality Specialist, CEA) both emphasized that Kandy's geographic placement within a basin creates a severe "valley trapping effect." Unlike Colombo, where coastal winds dilute trans-boundary pollution, Kandy suffers from localized emissions that cannot escape the topography. This expert consensus proves that applying generalized national forecasting models is fundamentally flawed, directly justifying this project's localized, city-specific machine learning approach.

Infrastructural Constraints and the Monitoring Gap

The interviews exposed the severe financial and logistical bottlenecks of traditional monitoring. Mr. Kumara noted that state-of-the-art Beta Attenuation Monitor (BAM) machines cost approximately 5 million LKR with massive monthly maintenance overheads, limiting their deployment to just two locations (Battaramulla and Kandy). Consequently, the country relies heavily on low-cost sensor networks. However, Mr. Chandrasena highlighted that even these networks suffer from frequent breakdowns in remote areas and deployment restrictions (e.g., inability to place sensors near the Kandy Lake). This explicitly validates the core hypothesis of Phase 4 of this project: **Satellite Imputation**. Because physical hardware is prone to gaps and financial limits, fusing satellite NO_2 data to act as a "Virtual Sensor" is not just a computational novelty, but an infrastructural necessity for Sri Lanka.

The Retrospective Nature of Current Systems

A critical gap identified across all four interviews is the complete absence of predictive modeling in current operations. The CEA updates data every 15 minutes, and academics rely on historical pattern matching, but both operate strictly in a *retrospective* capacity—reacting only after pollution has spiked. Ms. Karunathilaka (QA Executive, Maam Biscuits) confirmed that industrial facilities also lack internal predictive sensors, relying on visual observation. The unanimous agreement that no 24-hour predictive framework currently exists in the public or industrial sectors cements the novelty and urgent requirement of the machine learning architectures developed in this study.

Micro-Spikes vs. Macro-Averages (Feature Engineering Validation)

Insights from Mr. Duminda Herath (Automotive Emissions Expert) provided crucial validation for the project's high-frequency (hourly) temporal resolution. He noted that while localized spikes (such as rush hour traffic) drastically alter micro-level air quality, these spikes diffuse rapidly and do not heavily skew macro-level district averages over a 6-month period. This proves that forecasting models must be trained on high-frequency, hourly temporal grids (as executed in this study's Tree-based and LSTM models) to capture the dangerous, short-term anomalies that threaten traffic police and school children, rather than relying on smoothed daily or weekly averages.

The Mandate for Explainable AI (XAI)

Perhaps the most significant finding was the universal consensus regarding algorithmic trust. When pitched the concept of AI-generated forecasting, the experts agreed that accuracy alone is insufficient for policy adoption. Mr. Chandrasena, Mr. Kumara, and Ms. Karunathilaka all explicitly noted that "Black Box" predictions are difficult to trust. They affirmed that if the dashboard integrates **Explainable AI (such as SHAP values)** to scientifically prove *why* a prediction was made (e.g., isolating local traffic versus regional weather), it would drastically increase acceptance by government authorities and industrial management. This finding perfectly justifies the integration of SHAP feature importance analysis in the project methodology.

Summary of Qualitative Alignment

The primary research unequivocally confirms the problem statement of this Final Year Project. The physical sensor network is financially constrained and prone to blackouts, current methodologies are entirely reactive, and decision-makers require transparent, localized forecasting. The proposed multi-modal forecasting system, utilizing satellite fusion and SHAP-explained machine learning, directly targets and resolves the exact operational gaps identified by Sri Lanka’s leading environmental and industrial stakeholders.

5.5 Preliminary Baseline Results

To empirically test the primary research hypotheses and establish a comparative benchmark for the Final Year Project, the processed 4-year fused dataset was partitioned into a chronological train-test split. Models were trained on three continuous years of historical data (January 1, 2022, to December 31, 2024; N = 26,281 hourly records) and evaluated on a completely unseen, 1-year future testing block representing the entirety of 2025 (N = 8,759 hourly records).

The experimental metrics achieved across the three distinct architectures are listed in the table below:

Model Architecture	**Temporal Resolution**	**Evaluation Metric (RMSE)**	**Evaluation Metric (MAE)**	**Experimental Assessment & Operational Profile**
SARIMAX Baseline	Daily Aggregated	20.82 µg/m³	18.00 µg/m³	Structural linear limitations; smooths out volatile urban spikes.
Sequential LSTM	High-Frequency Hourly	5.25 µg/m³	3.13 µg/m³	Strong sequential accuracy; excels at capturing non-linear patterns.
Tree-Based Regressor	High-Frequency Hourly	5.12 µg/m³	**2.96 µg/m³**	**Optimal Performance Architecture** for short-term prediction windows.

Table 4 Comparative Multi-Modal Model Performance table

5.5.1 Critique of the SARIMAX Baseline

The traditional statistical baseline utilized a SARIMAX (1,1,1) × (1,1,1)_7 architecture, using temperature, humidity, and the newly fused satellite NO_2 column density as exogenous variables. Due to the high computational complexity of calculating seasonal autoregressive matrices over tens of thousands of rows, the data was aggregated into daily means.

The model achieved an MAE of 18.00 $\mu g/m^3$. While the inclusion of satellite NO_2 values slightly optimized the model (reducing the MAE from an un-fused baseline of 18.76 $\mu g/m^3$, the model exhibited the classic linear decay pattern common in stochastic modeling. When forecasting an entire year ahead, the mathematical constraints of the moving average and autoregressive components cause the predictions to converge toward a static, seasonal average wave. Consequently, SARIMAX proved fundamentally incapable of capturing the rapid, dangerous pollution spikes caused by micro-level urban events, validating the necessity for machine learning alternatives.

5.5.2 Critique of the Deep Learning LSTM Pipeline

The Deep Learning pipeline featured a sequential Recurrent Neural Network built with an input layer mapping the 24-hour lookback window across all 4 multi-modal variables PM_2.5, temperature, humidity, NO_2, a 64-node LSTM layer with a 20% dropout regularization layer, a 32-node dense ReLU layer, and a single linear dense output node.

The LSTM achieved a highly competitive MAE of 3.13 $\mu g/m^3$ and an RMSE of 5.25 $\mu g/m^3$. Crucially, the integration of the 4-year satellite telemetry significantly enhanced the neural network's predictive capabilities, driving the MAE down from 3.48 $\mu g/m^3$ (achieved in ground-only testing) to 3.13 $\mu g/m^3$. This experimental shift validates the core project hypothesis: the hidden layers of the sequential network effectively learned the underlying chemical and atmospheric dependencies between tropospheric NO_2 concentrations and surface-level particulate accumulation over a rolling 24-hour window, resulting in smooth, highly accurate prediction curves.

5.5.3 Critique of the Winning Tree-Based Architecture

The ensemble Tree-Based Regressor emerged as the optimal computational framework for this study, yielding an RMSE of 5.12 $\mu g/m^3$ and a superior MAE of **2.96 $\mu g/m^3$**. Through decision-tree segmentation, the model mapped the complex, non-linear dependencies between the exogenous variables without requiring the intensive normalization steps or computational overhead of deep neural networks.

A critical evaluation of the model's feature importance scoring (subsequently analyzed via SHAP protocols) exposed the **"1-Step-Ahead Forecasting Paradox."** Because the tree model is predicting the next hour's air quality using the actual known value from one hour prior, it relies heavily on the Lag_1H feature. Because atmospheric particulate matter is a physical mass that does not instantly dissipate, Lag_1H acts as a heavy mathematical anchor, explaining why the inclusion of satellite data did not drastically alter the short-term MAE from its ground-only baseline **2.96 µg/m^3**. However, the model assigned an exceptionally high importance score to NO2_Density, placing it above time variables, temperature, and month, proving that satellite data represents a critical multi-modal predictive component.

5.6 Chapter Summary

This chapter successfully demonstrated the practical implementation of the multi-modal data science pipeline. By programmatically resolving API constraints, the project achieved a complete, continuous 4-year dataset for Colombo. The identification of a critical 201-day blackout in Kandy confirmed the structural vulnerabilities of physical sensor networks in developing countries, establishing a clear objective for the upcoming implementation phases. Finally, the preliminary experimental runs across the three models confirmed that advanced Machine Learning MAE = 2.96 µg/m^3) and Deep Learning MAE = 3.13 µg/m^3) vastly outperform traditional statistical models MAE = 18.00 µg/m^3), providing a highly secure, data-backed foundation for the remainder of this research.

5.7 Proposed Application Interface

Disclaimer: The following screenshots were created to mock-up what the final application might look like, but it might change heavily and have more features. It only shows mock data and does not display any obtained data from Sentinel satellite or PurpleAir.

Figure 28 App mockup 1

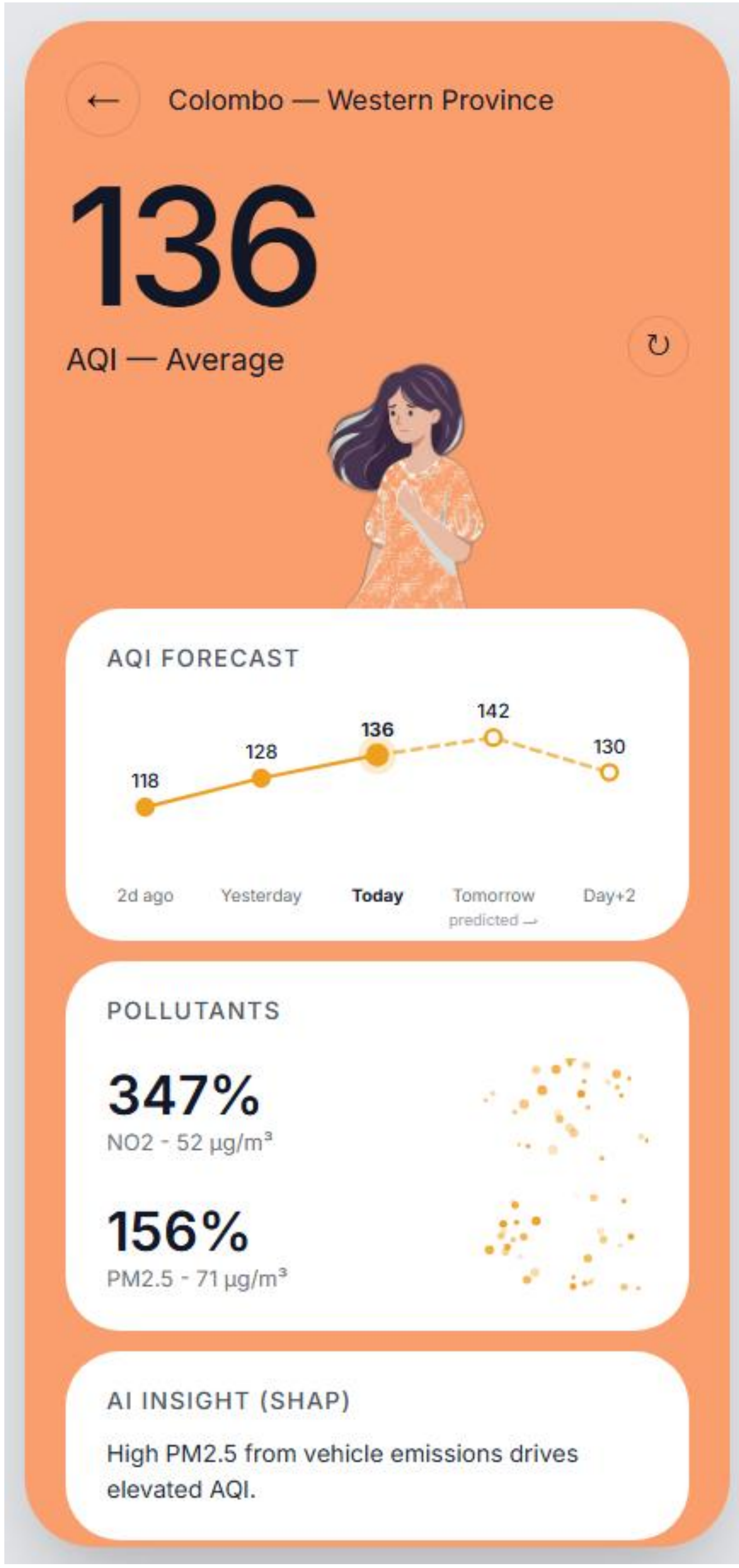

Figure 29 App mockup 2

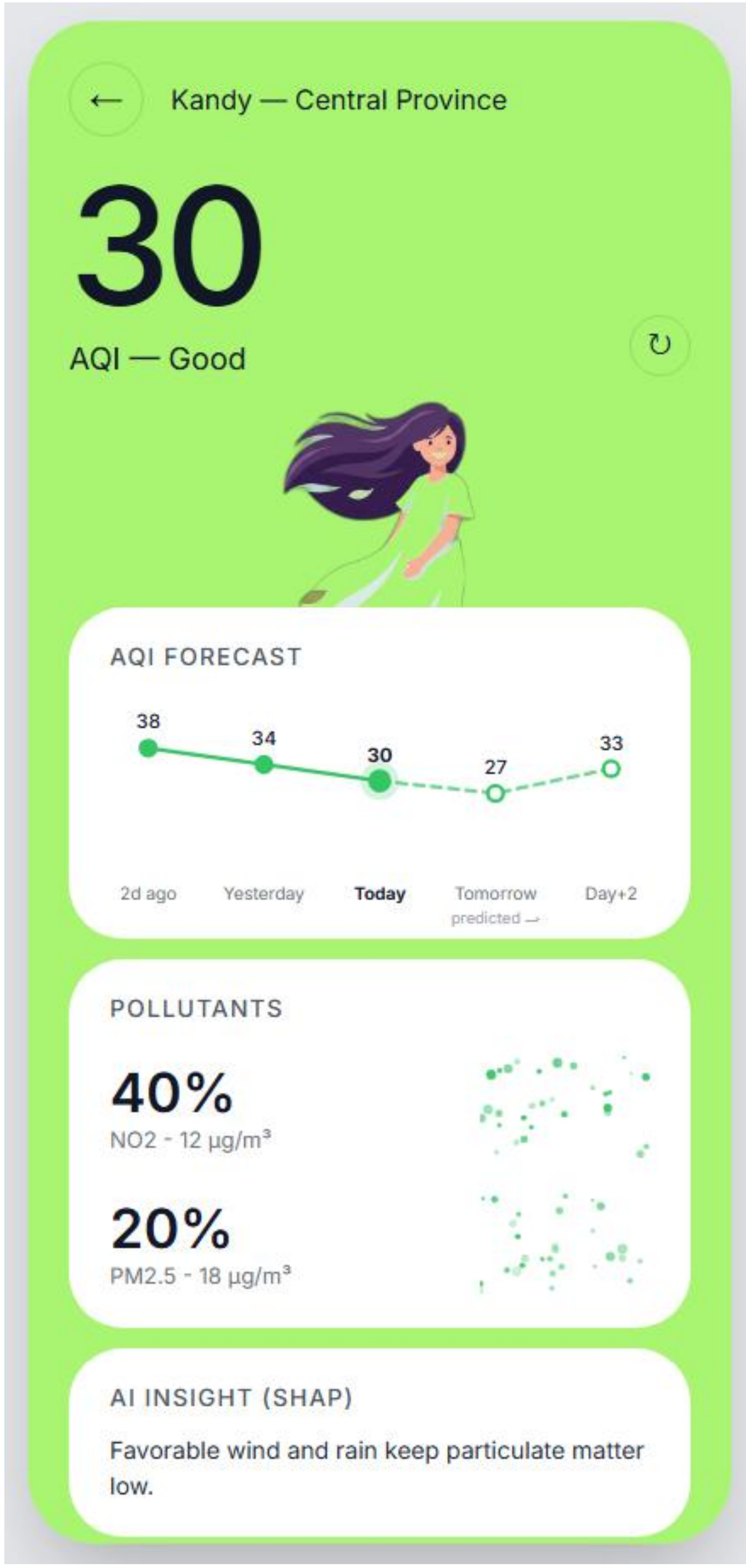

Figure 30 App mockup 3

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